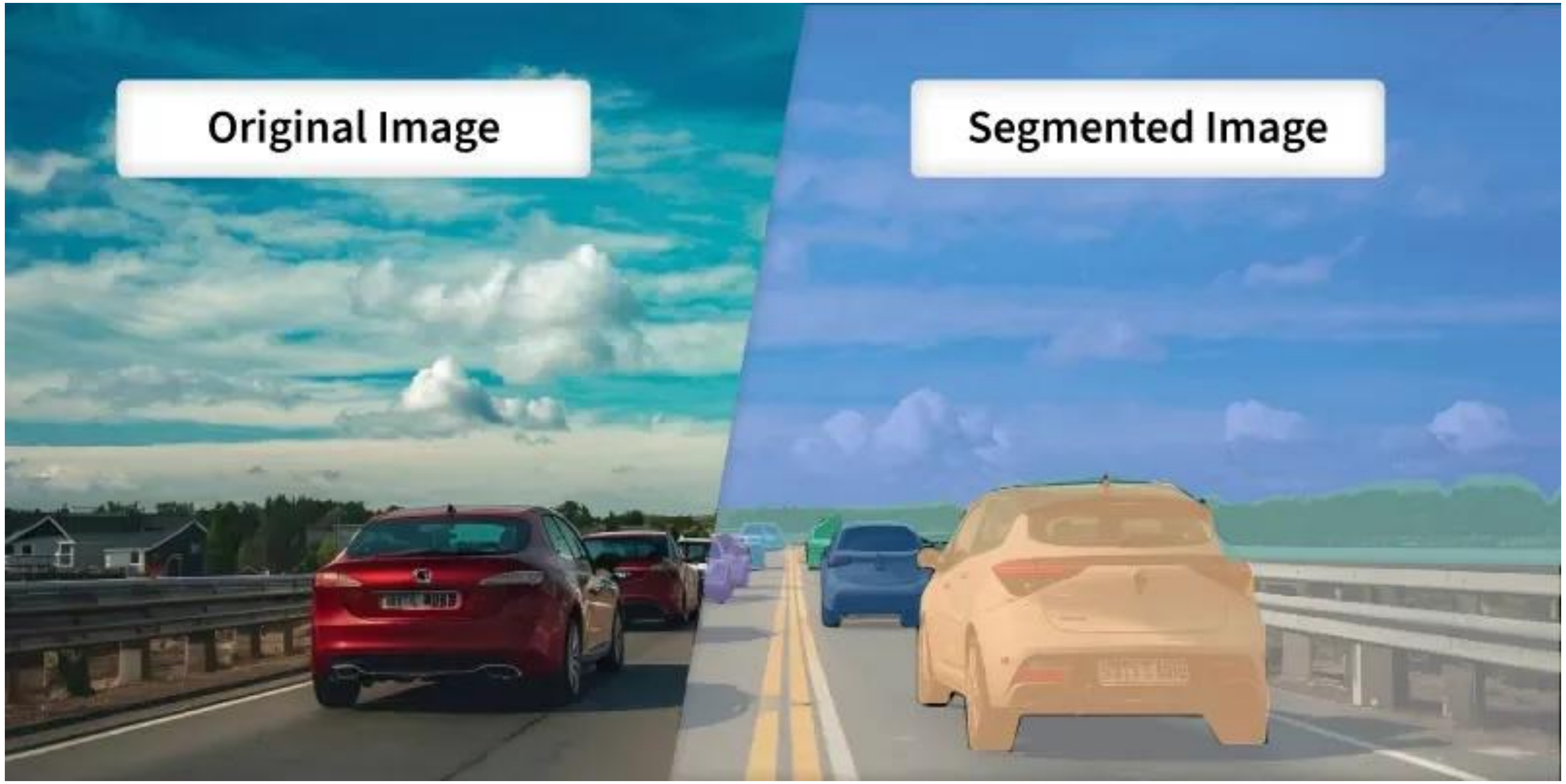


Image Segmentation Tutorial

ZJU GIS 2026 Summer

Presenter: Zhenyuan Chen

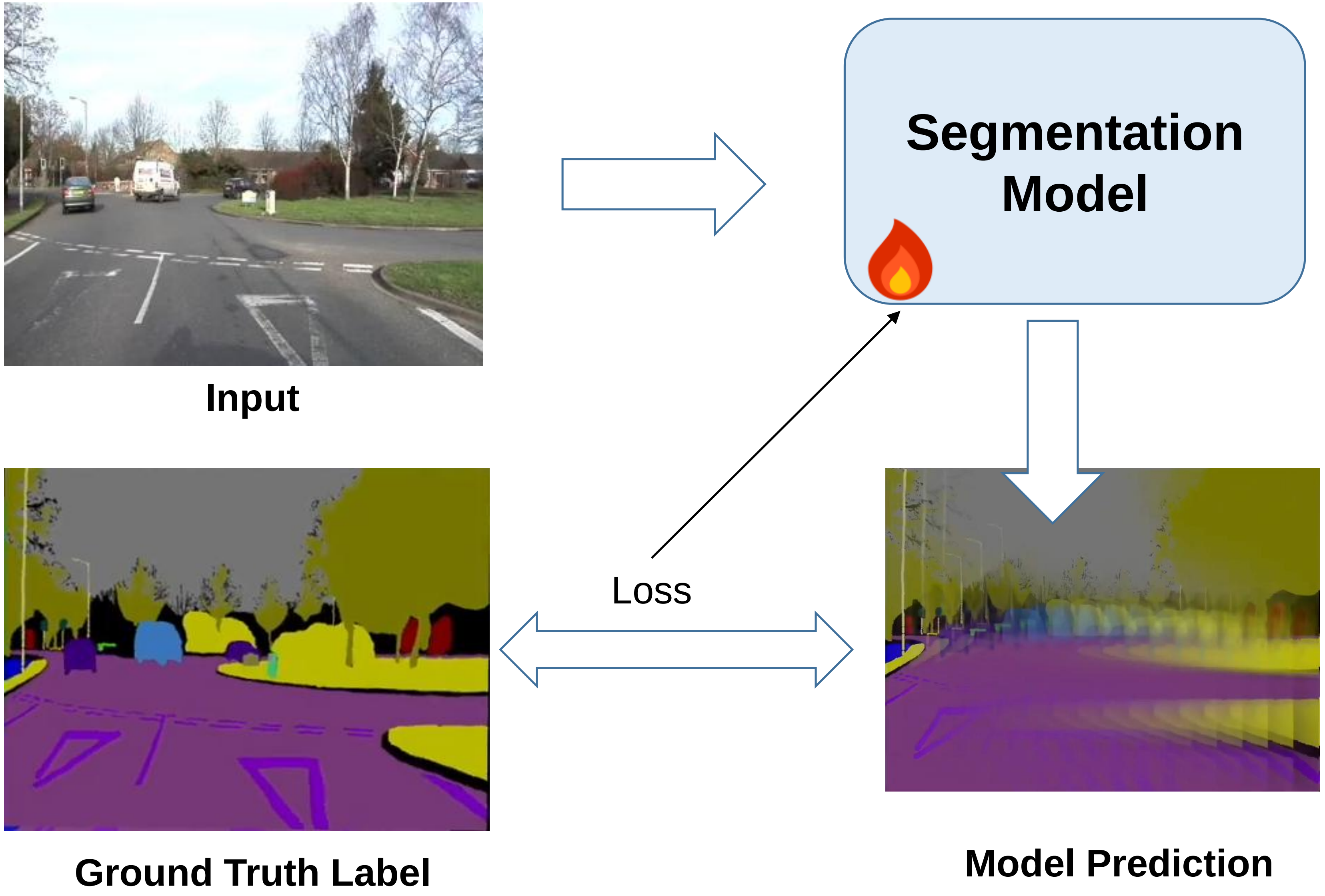
Outline

1. Conventional Image Segmentation
   - **Semantic Segmentation**
   - Instance & Panoptic Segmentation
2. Foundation Model Era
   - SSL Backbone + Task Head
   - CLIP based
   - Diffusion Model based
   - **MLLM based**
   - SAM Family
3. Case Study in Cropland Recognition

Zhou et al., Image Segmentation in Foundation Model Era: A Survey, arXiv 2024

Conventional Image Segmentation

Preliminary



- Dataset
- Model Architecture
- Training Loss
- Evaluation

Preliminary

Input: An RGB image with shape as $\mathbb{R}^{H\times W\times 3}$

Target: A label map containing K+1 different classes with shape as $S \in \{0,1,2,\ldots,K\}^{H\times W}$

Model Architecture: CNNs, Transformers, Mamba etc.

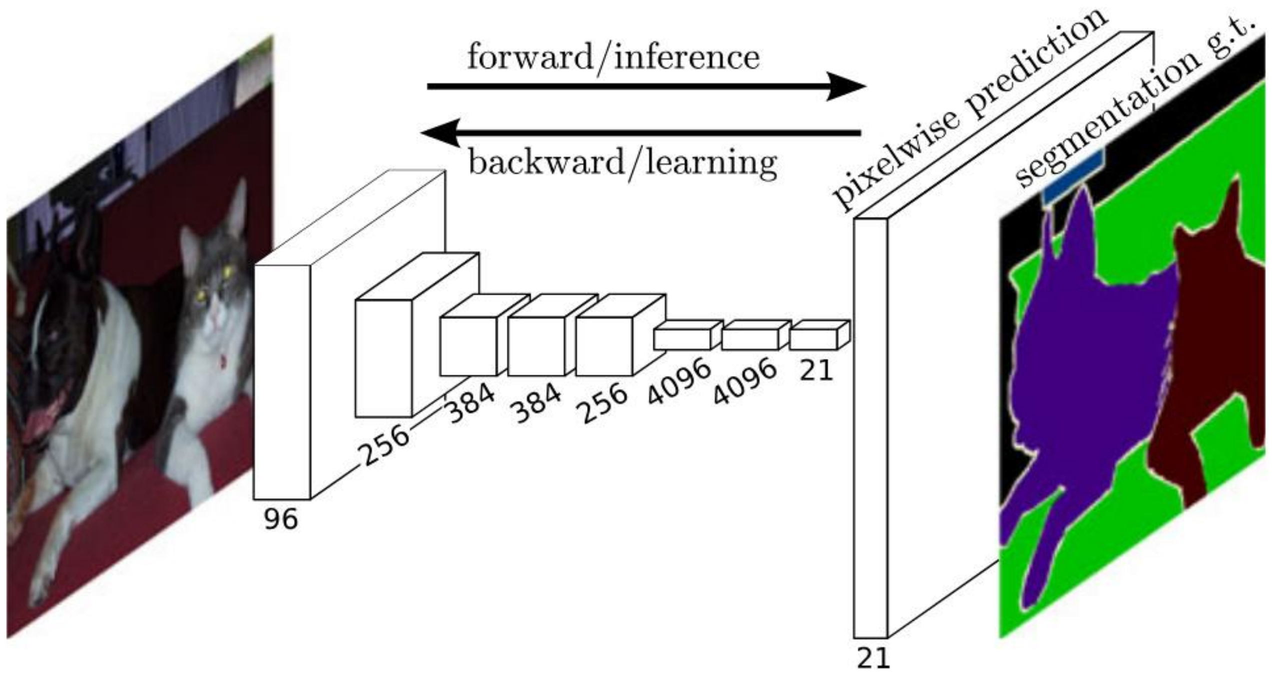
Loss: Cross-Entropy Loss, Weighted Cross-Entropy Loss, Dice Loss, Focal Loss, Lovasz Loss

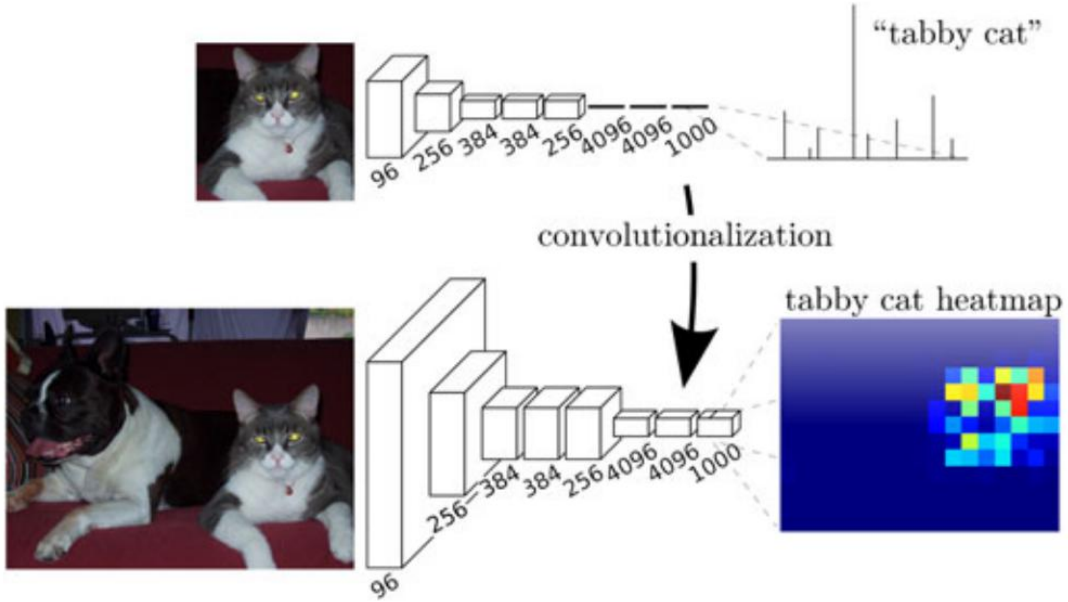
Metrics: Intersection over Union (IoU), Mean IoU (mIoU), Average Precision (AP), Mean AP (mAP)**,** F1

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log(\hat{y}_{i,c})$$

N = Total number of pixels in the image/batch.
C = Number of classes.
$y_{i,c}$ = Ground truth (1 if pixel i belongs to class c, 0 otherwise).
$\hat{y}_{i,c}$ = Predicted probability for pixel i belonging to class c (after Softmax).

Semantic Segmentation: FCNs

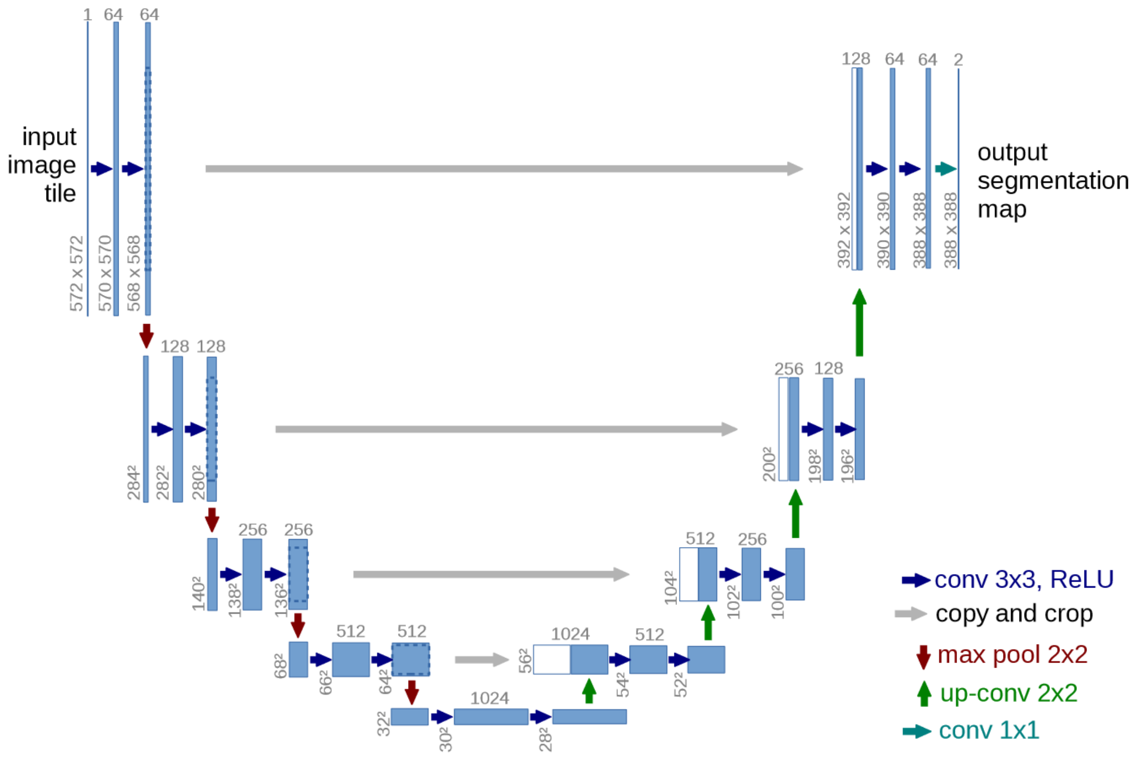


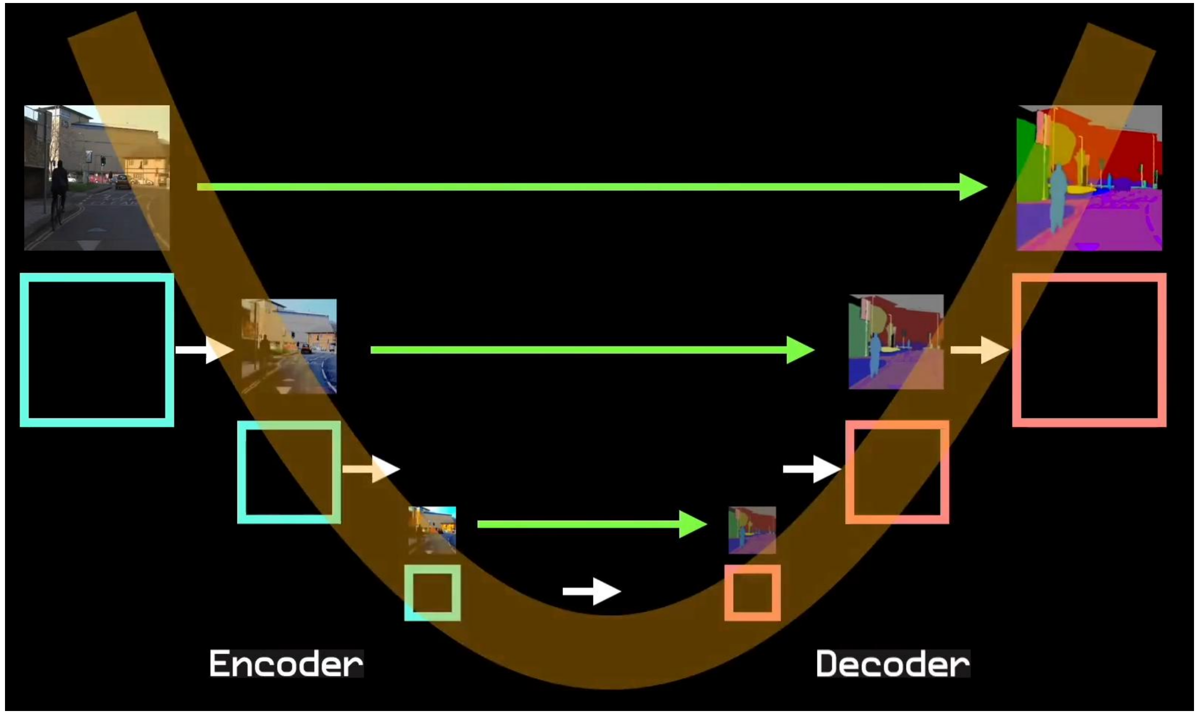


Left: Overview of Fully Convolutional Networks (FCNs) for image segmentation. Right: FCNs are built upon image classification models (e.g. AlexNet, VGG, GoogLeNet) by transforming fully connected layers into convolution layers enables a classification net to output a spatial map.

Long et al., Fully Convolutional Networks for Semantic Segmentation, CVPR 2015
Shelhamer et al., Fully Convolutional Networks for Semantic Segmentation, TPAMI 2017

Semantic Segmentation: U-Net

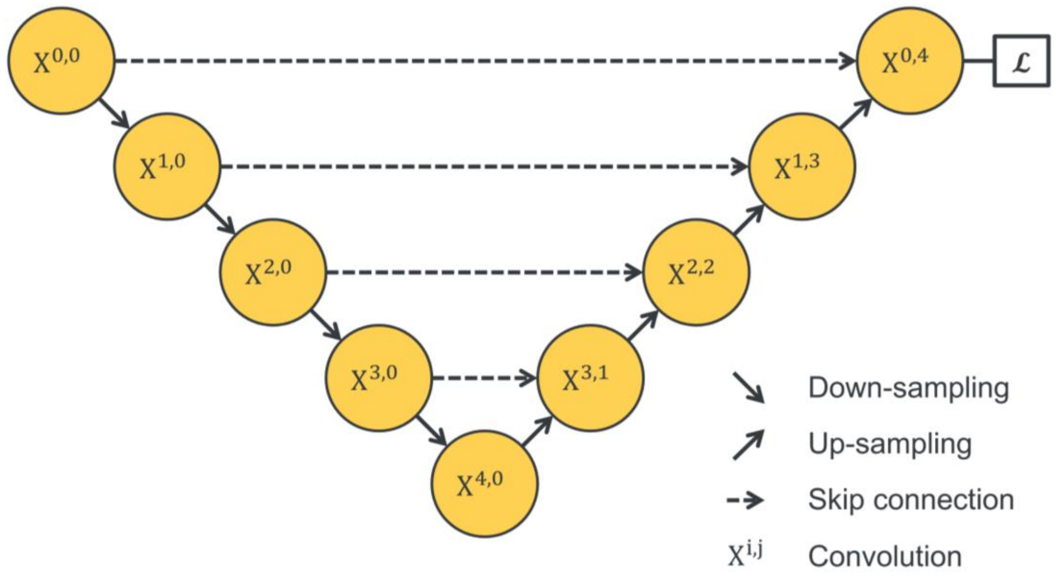




Left: Detailed architecture of U-Net. Right: High-level illustration of U-Net.

Ronneberger et al., U-Net: Convolutional Networks for Biomedical Image Segmentation, MICCAI 2015

Semantic Segmentation: U-Net



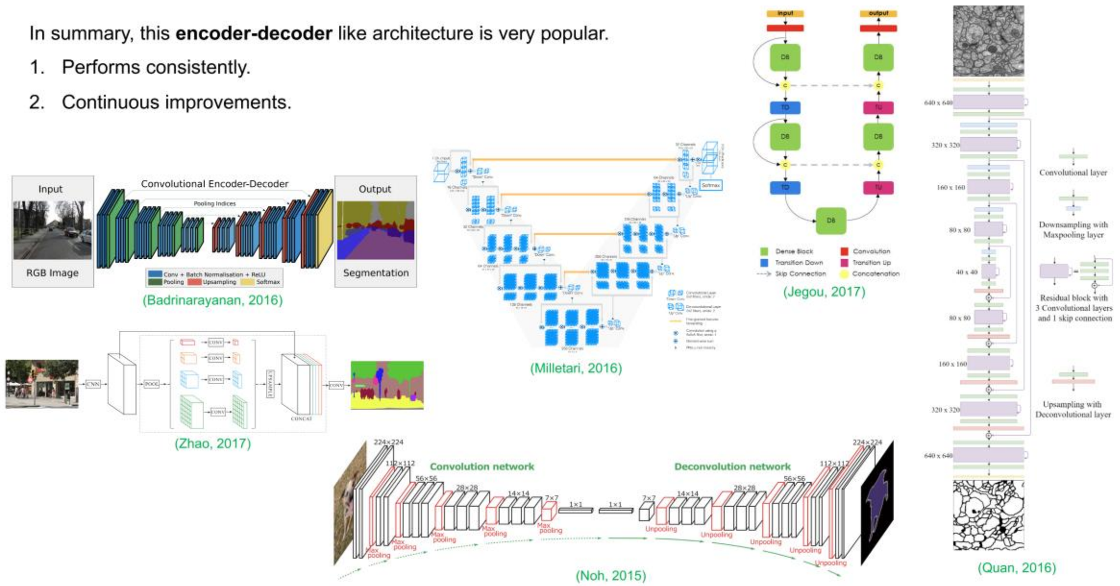


Image Credit: UNet++: A Nested U-Net Architecture for Medical Image Segmentation

The **feature extractor** is important:

1. Good feature representation.
2. Fast convergence speed.

The **down-sampling** is important:

1. Robust against small input variance.
2. Reduce overfitting.
3. Reduce computation cost.
4. Enlarge receptive field area.

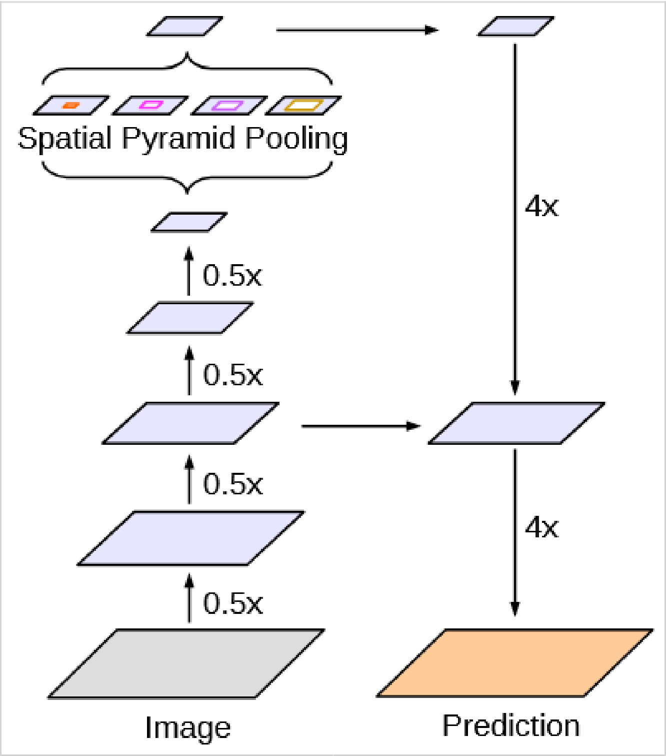
The **up-sampling** is important:

1. Recovers lost resolution in down-sampling
2. Guides encoder to select important information

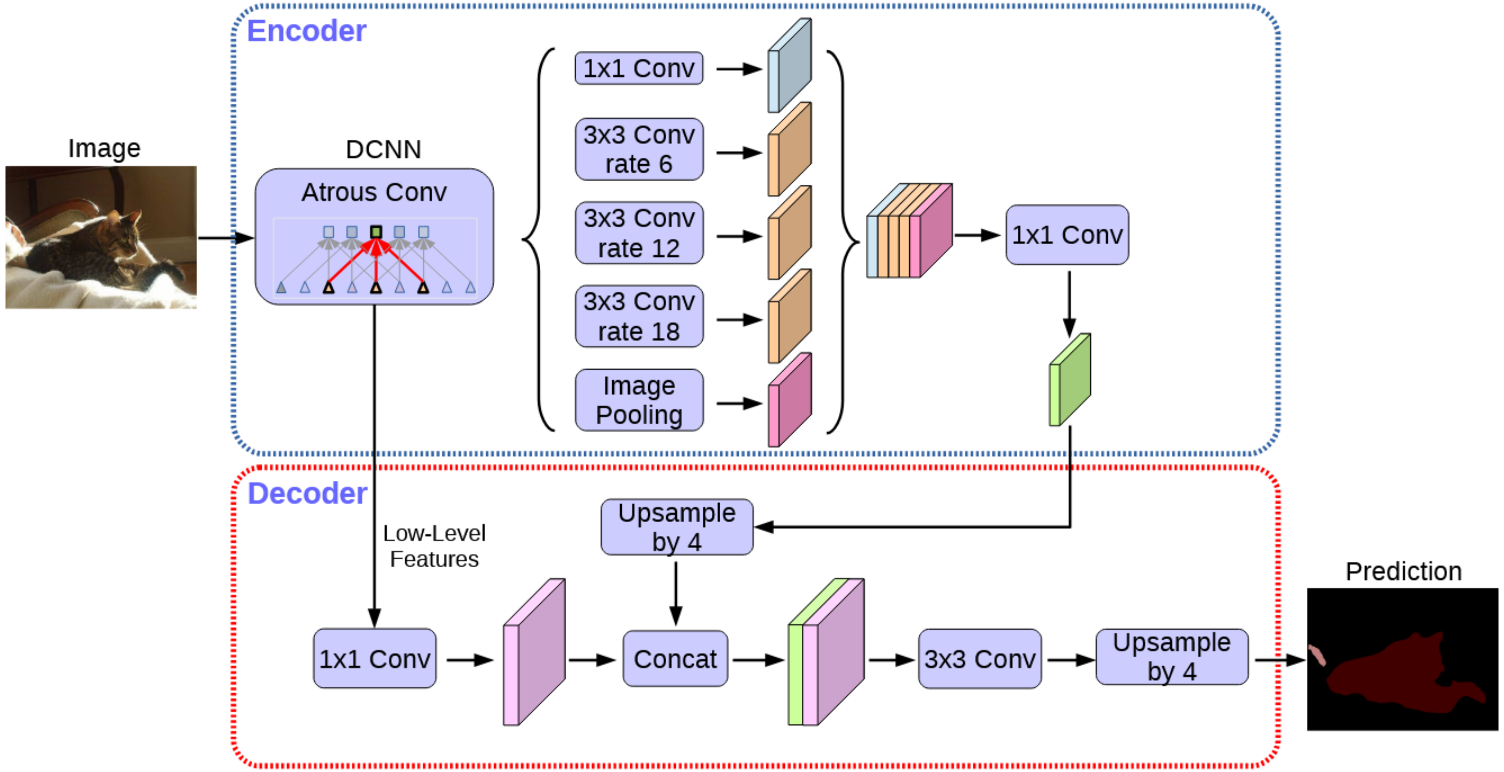
The **skip connection** is important:

1. Fights the vanishing gradient problem.
2. Learns pyramid level features.
3. Recover info loss in down-sampling.

Semantic Segmentation: DeeplabV3+



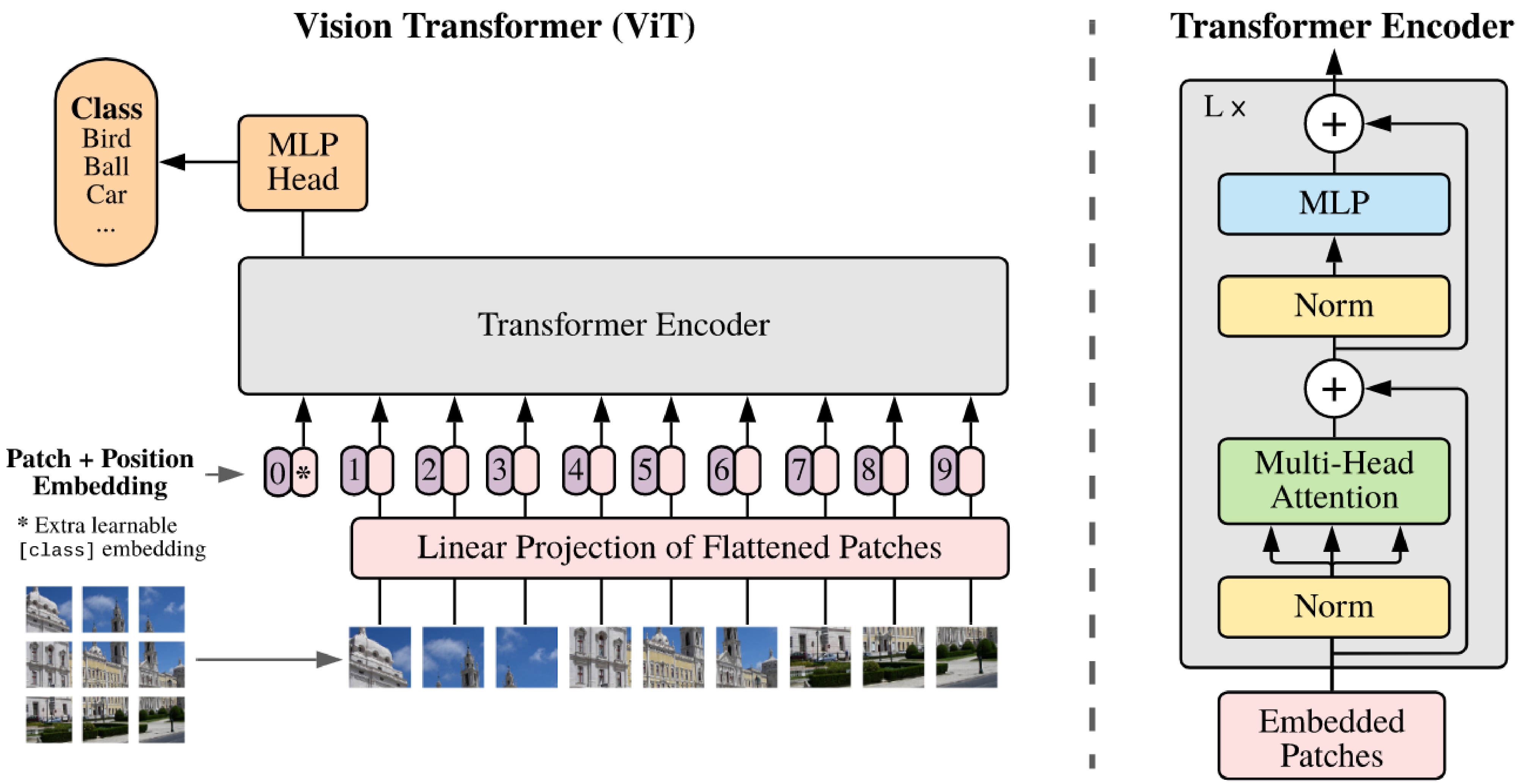
(c) Encoder-Decoder with Atrous Conv



Left: High-Level architecture of DeeplabV3+. Right: Detailed Architecture of DeeplabV3+. It adopts several advanced techniques including astrous (dilated) convolution and spatial pyramid features, achieving SOTA at that time.

Chen et al., Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation, ECCV 2018

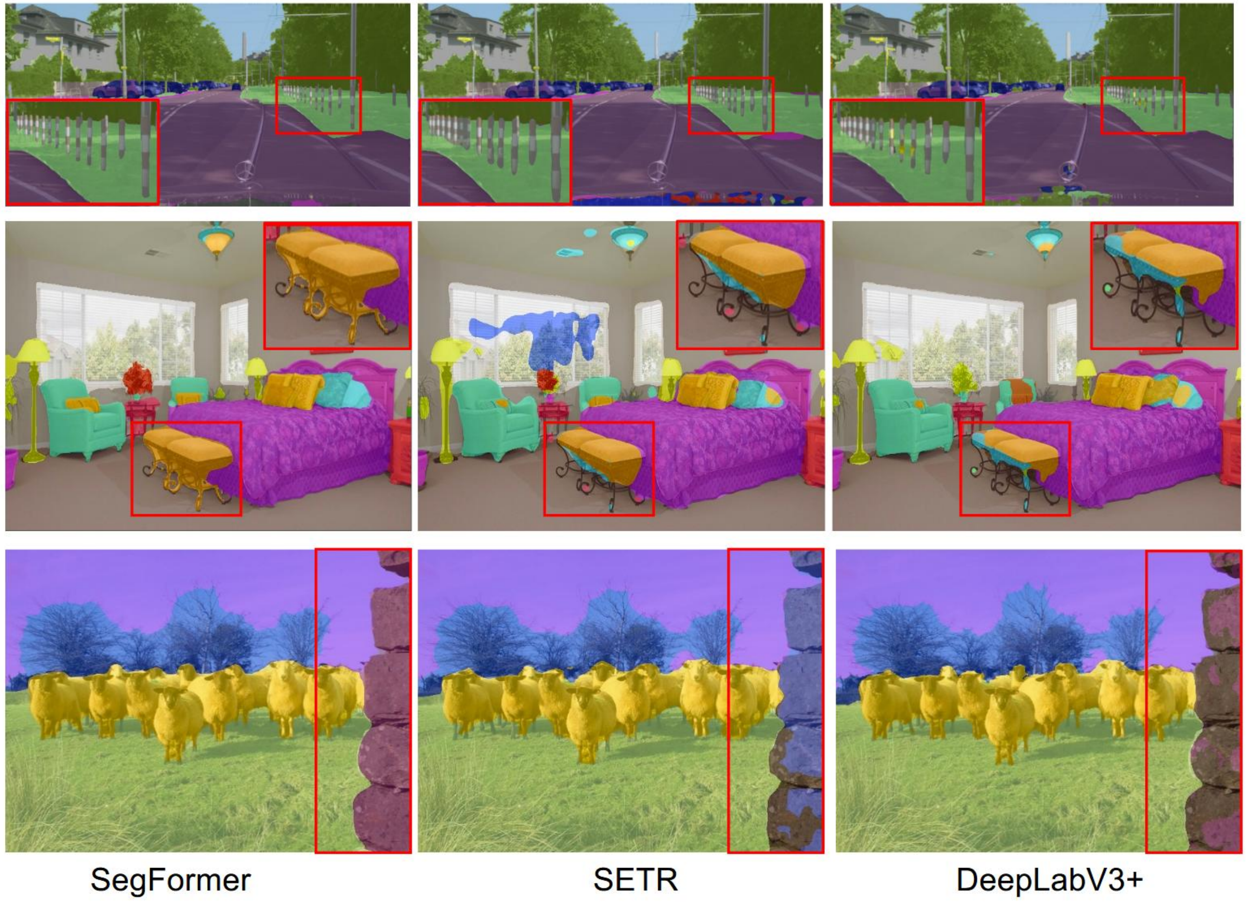
Preliminary: ViT

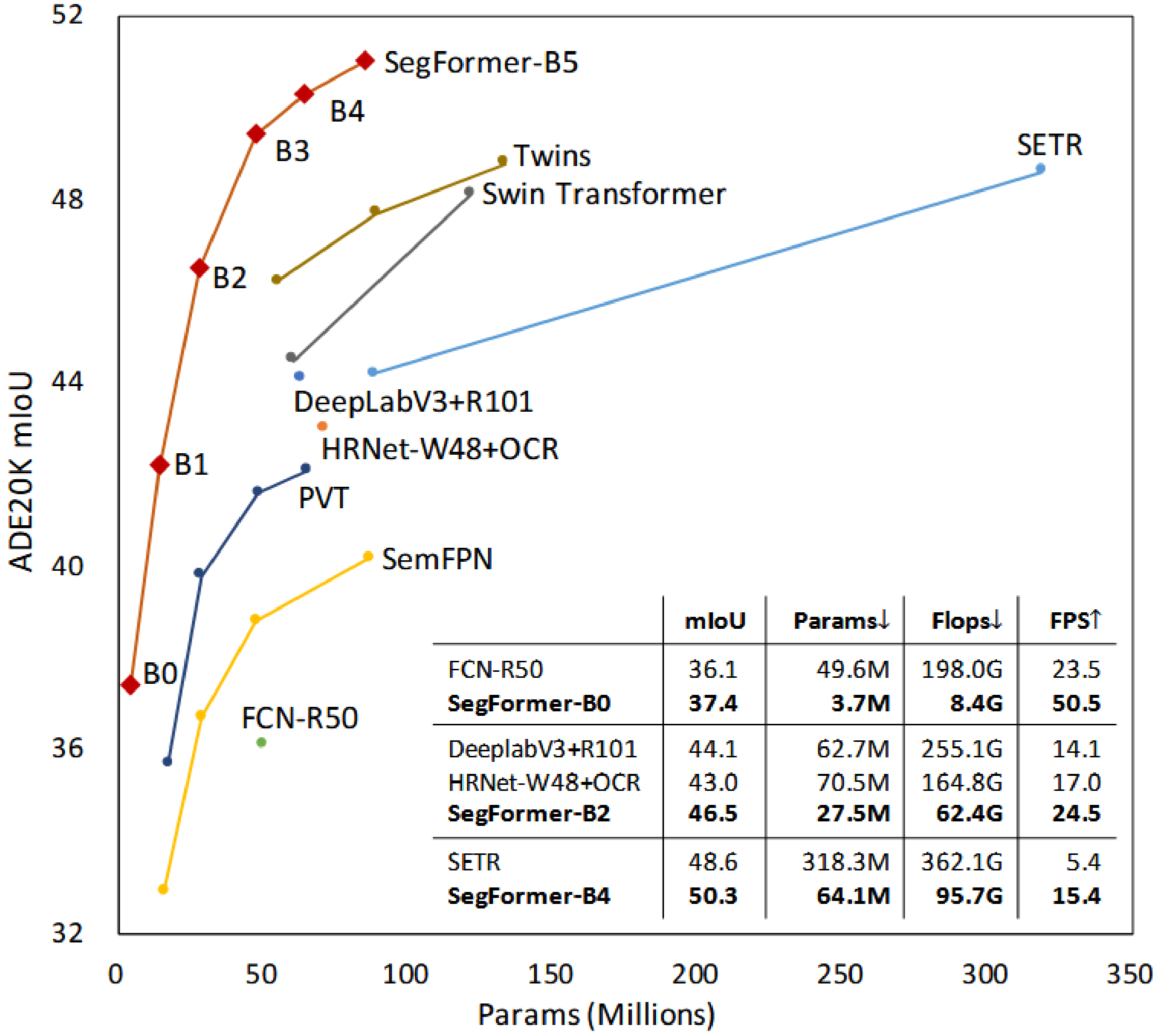


Model architecture of Vision Transformer (ViT). An image is firstly patchified and flattened as a sequence.

Alexey et al., An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, ICLR 2021

Semantic Segmentation: SegFormer



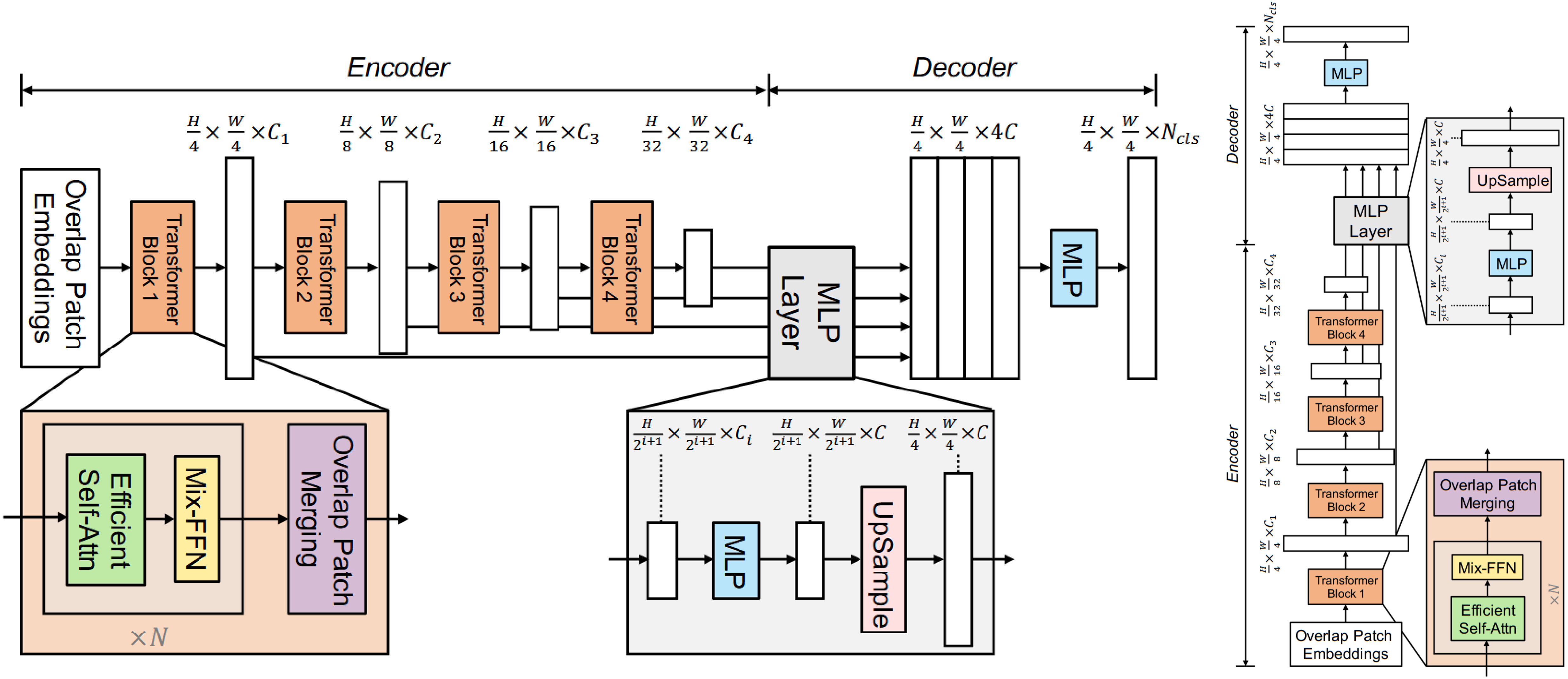


	mIoU	Params↓	Flops↓	FPS↑
FCN-R50	36.1	49.6M	198.0G	23.5
SegFormer-B0	**37.4**	**3.7M**	**8.4G**	**50.5**
DeeplabV3+R101	44.1	62.7M	255.1G	14.1
HRNet-W48+OCR	43.0	70.5M	164.8G	17.0
SegFormer-B2	**46.5**	**27.5M**	**62.4G**	**24.5**
SETR	48.6	318.3M	362.1G	5.4
SegFormer-B4	**50.3**	**64.1M**	**95.7G**	**15.4**

Left: Results of SegFormer compared with baselines on semantic segmentation. Right: Performance plot.

Xie et al., SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers, NeurIPS 2021

Semantic Segmentation: SegFormer



The proposed SegFormer framework consists of two main modules: A hierarchical Transformer encoder to extract coarse and fine features; and a lightweight All-MLP decoder to directly fuse these multi-level features and predict the semantic segmentation mask. “FFN” indicates feed-forward network

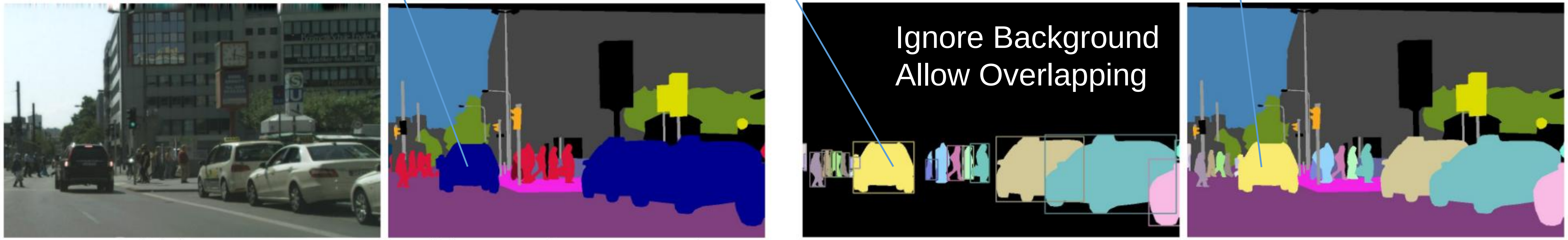
Xie et al., SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers, NeurIPS 2021

Semantic vs. Instance vs. Panoptic

Every pixel has a class label.

Every pixel has a class label and an instance id.

Every instance has an instance id.



(a) image (b) semantic segmentation (c) instance segmentation (d) panoptic segmentation

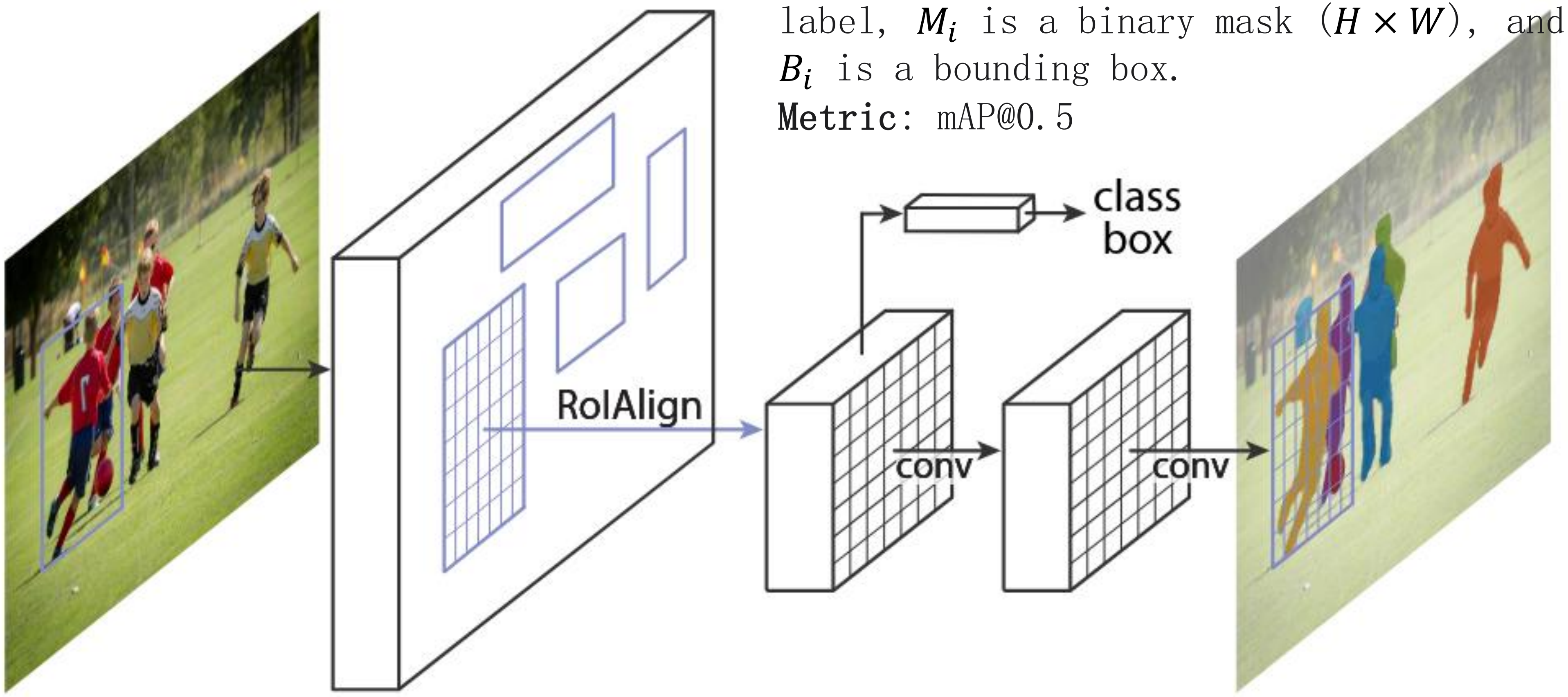
The Panoptic Segmentation task: (1) encompasses both **stuff** and **thing** classes, (2) uses a simple but general format, and (3) introduces a uniform evaluation metric for all classes. Panoptic segmentation generalizes both semantic and instance segmentation and we expect the unified task will present novel challenges and enable innovative new methods.

Kirillov et al., Panoptic Segmentation, CVPR 2019

Instance Segmentation: Mask R-CNN

Target: A set of N discrete tuples: $\{(c_i, M_i, B_i)\}_{i=1}^{N}$, where c_i is the class label, M_i is a binary mask $(H \times W)$, and B_i is a bounding box.

Metric: mAP@0.5



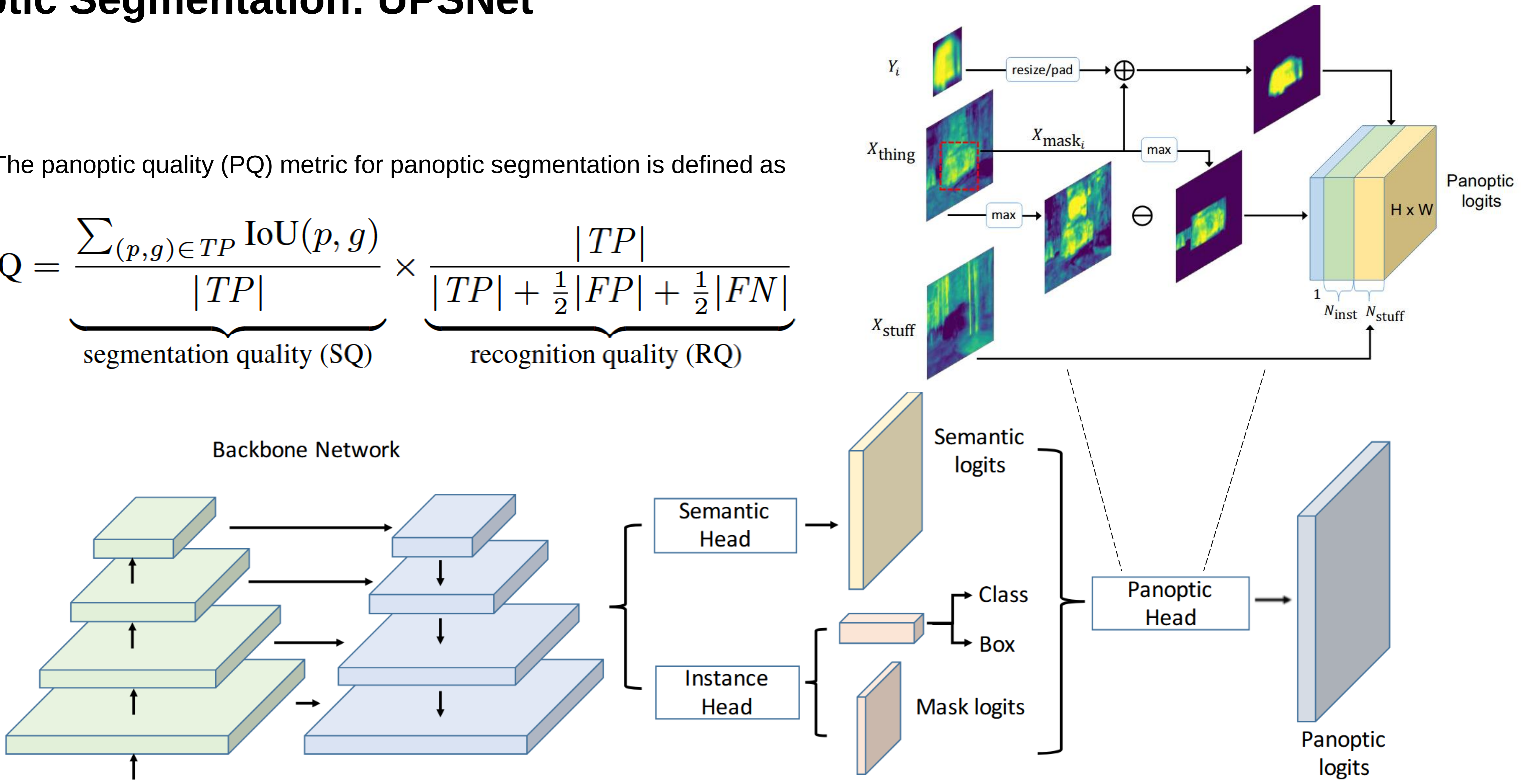
The Mask R-CNN framework for instance segmentation.

He et al., Mask R-CNN, ICCV 2017

Panoptic Segmentation: UPSNet

The panoptic quality (PQ) metric for panoptic segmentation is defined as

$$\mathrm{PQ} = \underbrace{\frac{\sum_{(p,g)\in TP} \mathrm{IoU}(p,g)}{|TP|}}_{\text{segmentation quality (SQ)}} \times \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\text{recognition quality (RQ)}}$$

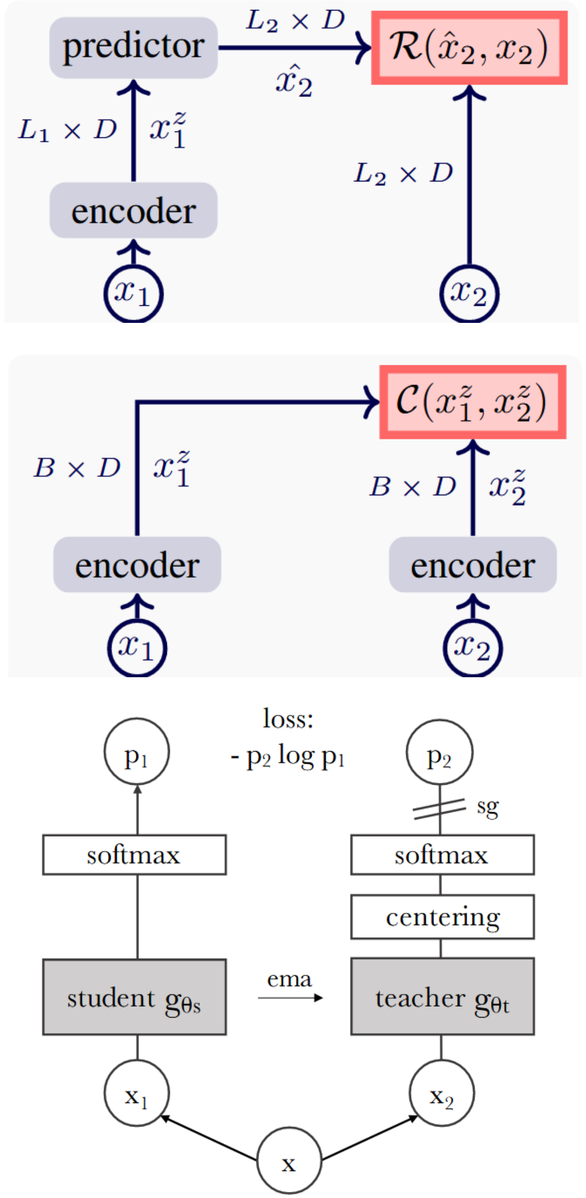


The UPSNet framework for Panoptic segmentation.

Xiong et al., UPSNet: A Unified Panoptic Segmentation Network, ICCV 2019

Foundation Model Era

SSL Backbone + Task Head



$\mathcal{R}$: reconstructive loss
e.g. mean squared error

$\mathcal{C}$: contrastive loss
e.g. InfoNCE

B : batch size

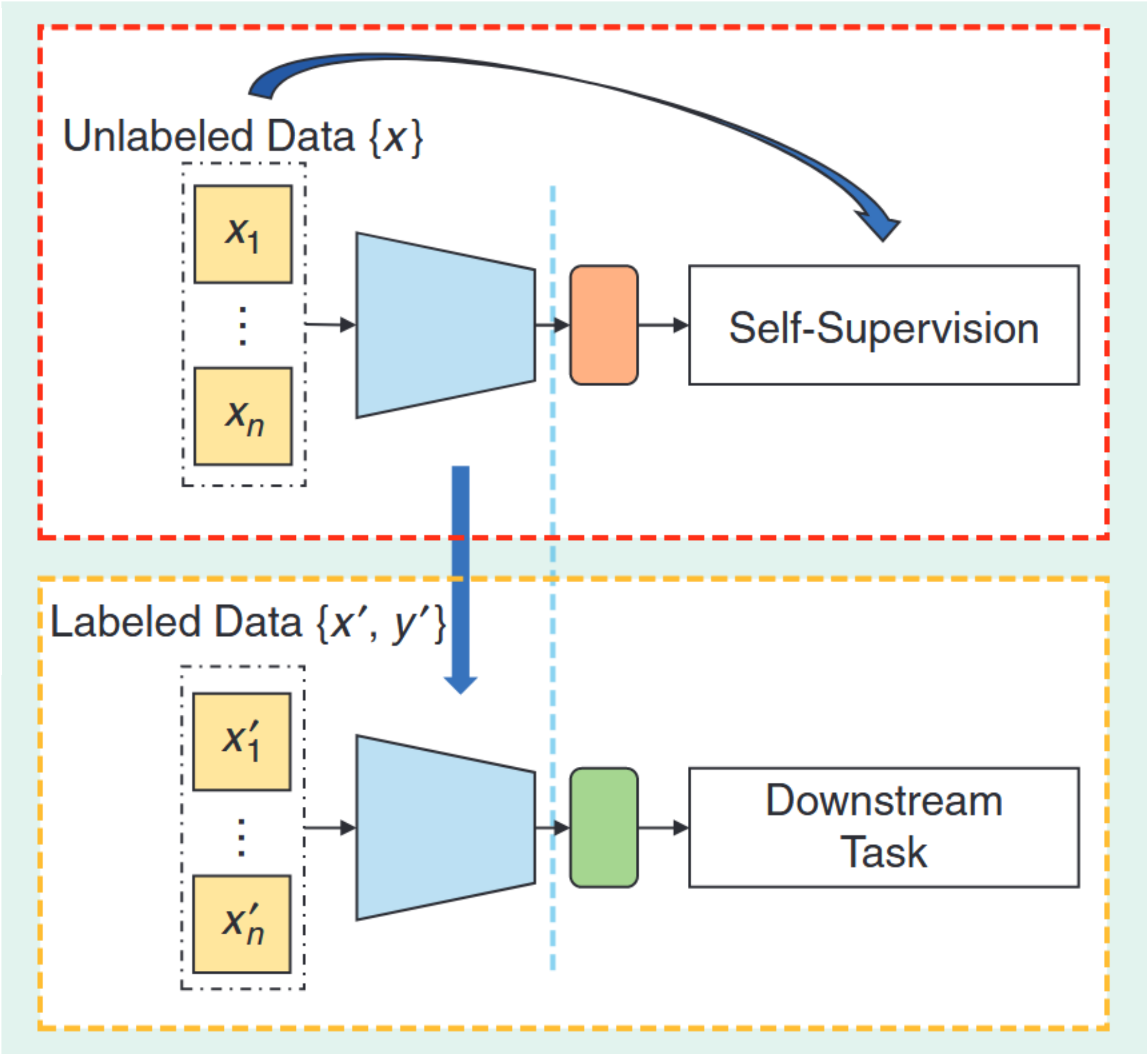
L : sequence length

D : feature dim

z : latent

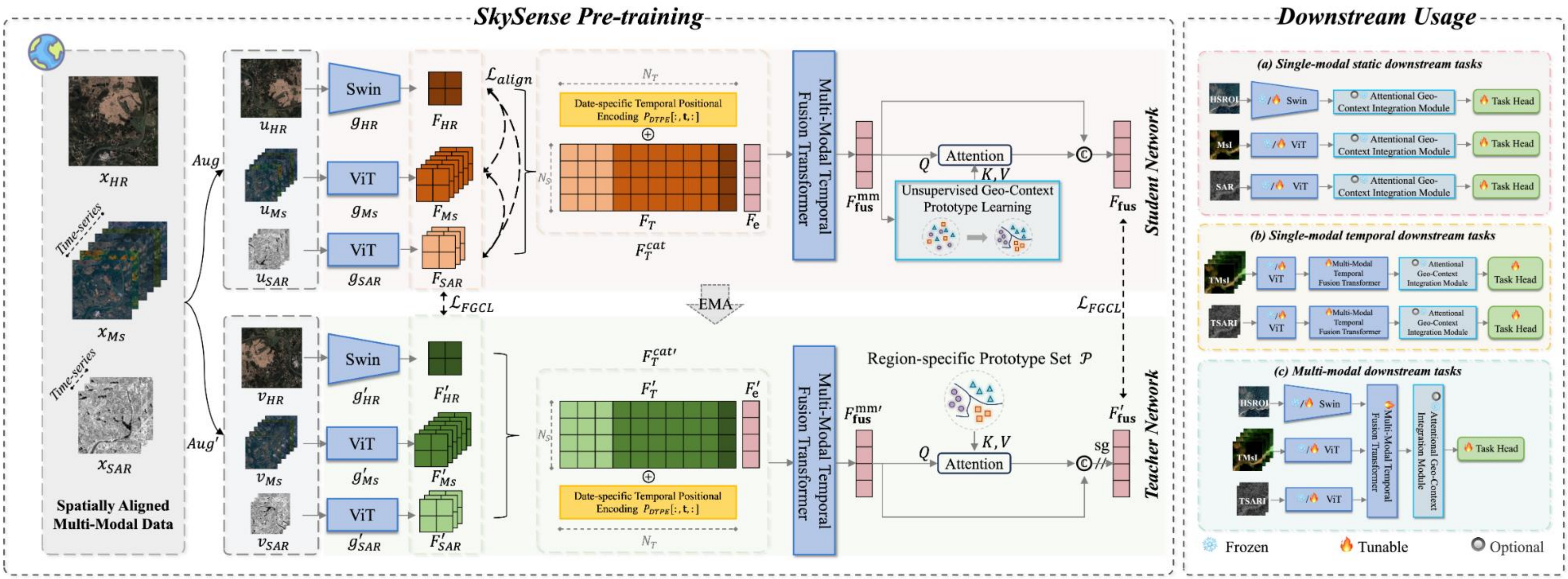
$1/2/3$: views

Three Representative SSL framework as MAE, SimCLR and DINO (from top to down).



Wang et al., Self-Supervised Learning in Remote Sensing: A Review. GRSM 2022
Tseng et al., Galileo: Learning Global & Local Features of Many Remote Sensing Modalities, ICML 2025

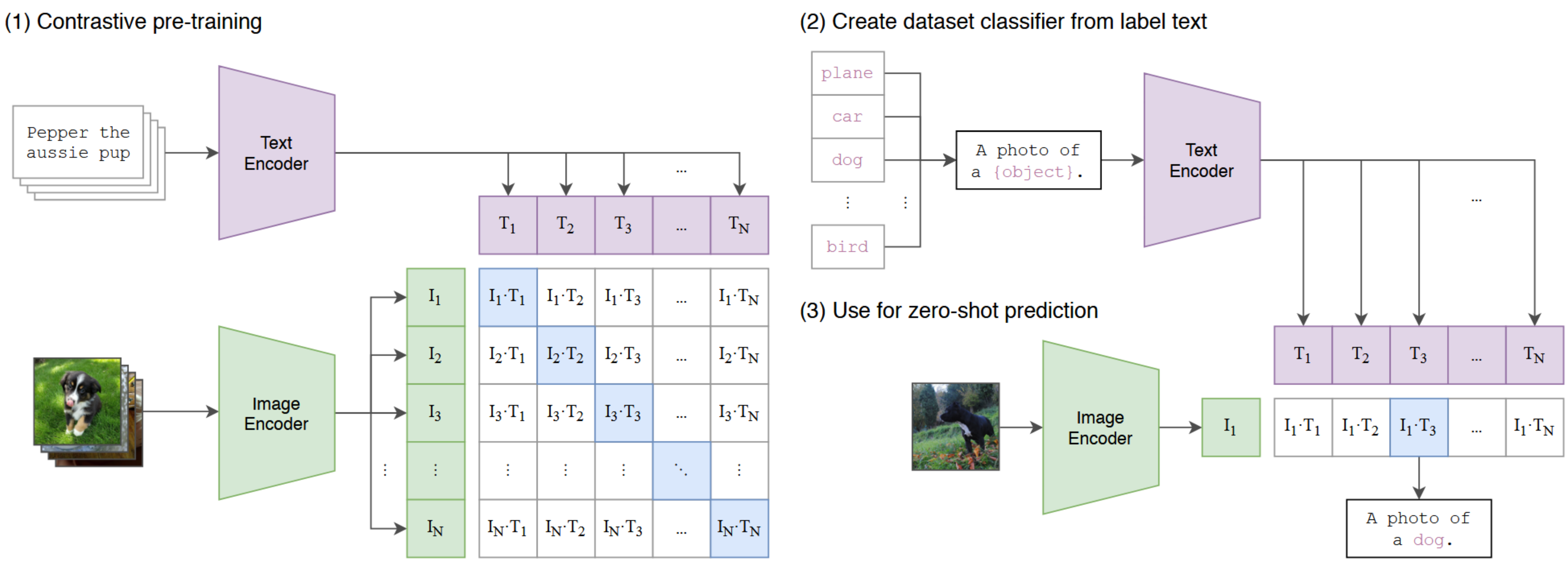
SSL Backbone + Task Head: SkySense



Overview of SkySense pre-training and downstream usage. The Self-supervised Learning (SSL) framework is borrowed from DINO.

Guo et al., SkySense: A Multi-Modal Remote Sensing Foundation Model Towards Universal Interpretation for Earth Observation Imagery, CVPR 2024
Caron et al., Emerging Properties in Self-Supervised Vision Transformers, ICCV 2021

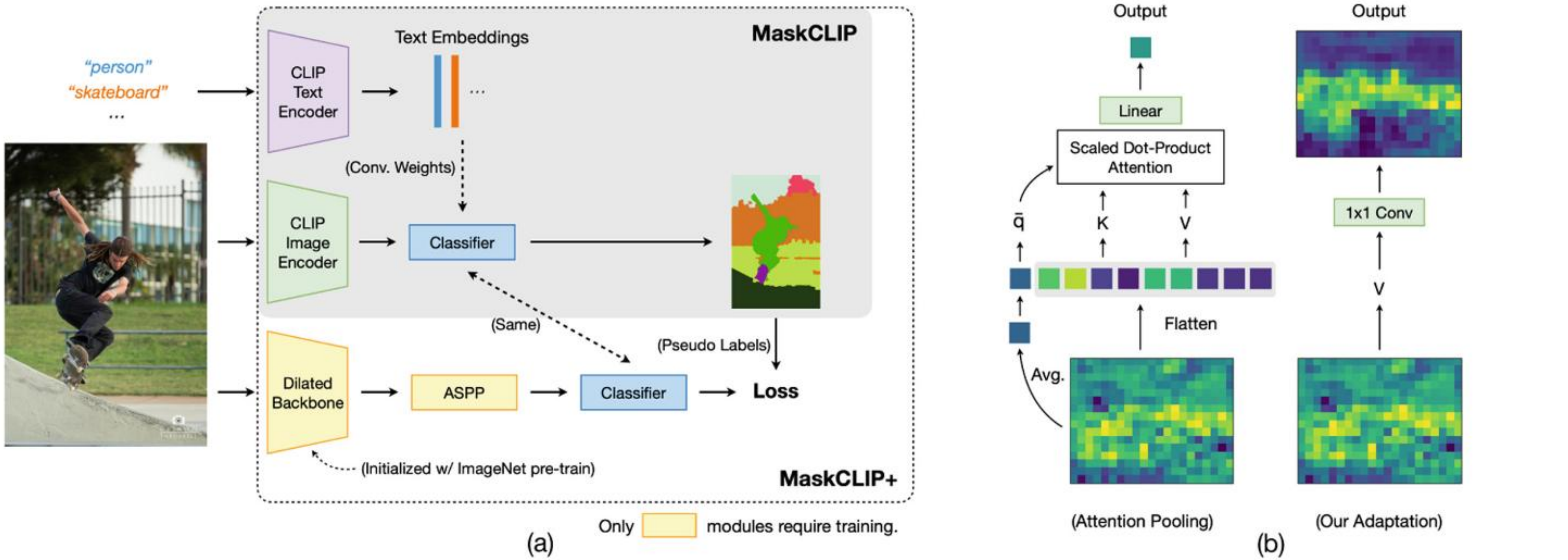
Preliminary: CLIP



Overview of CLIP.

Radford et al., Learning Transferable Visual Models From Natural Language Supervision, ICML 2021

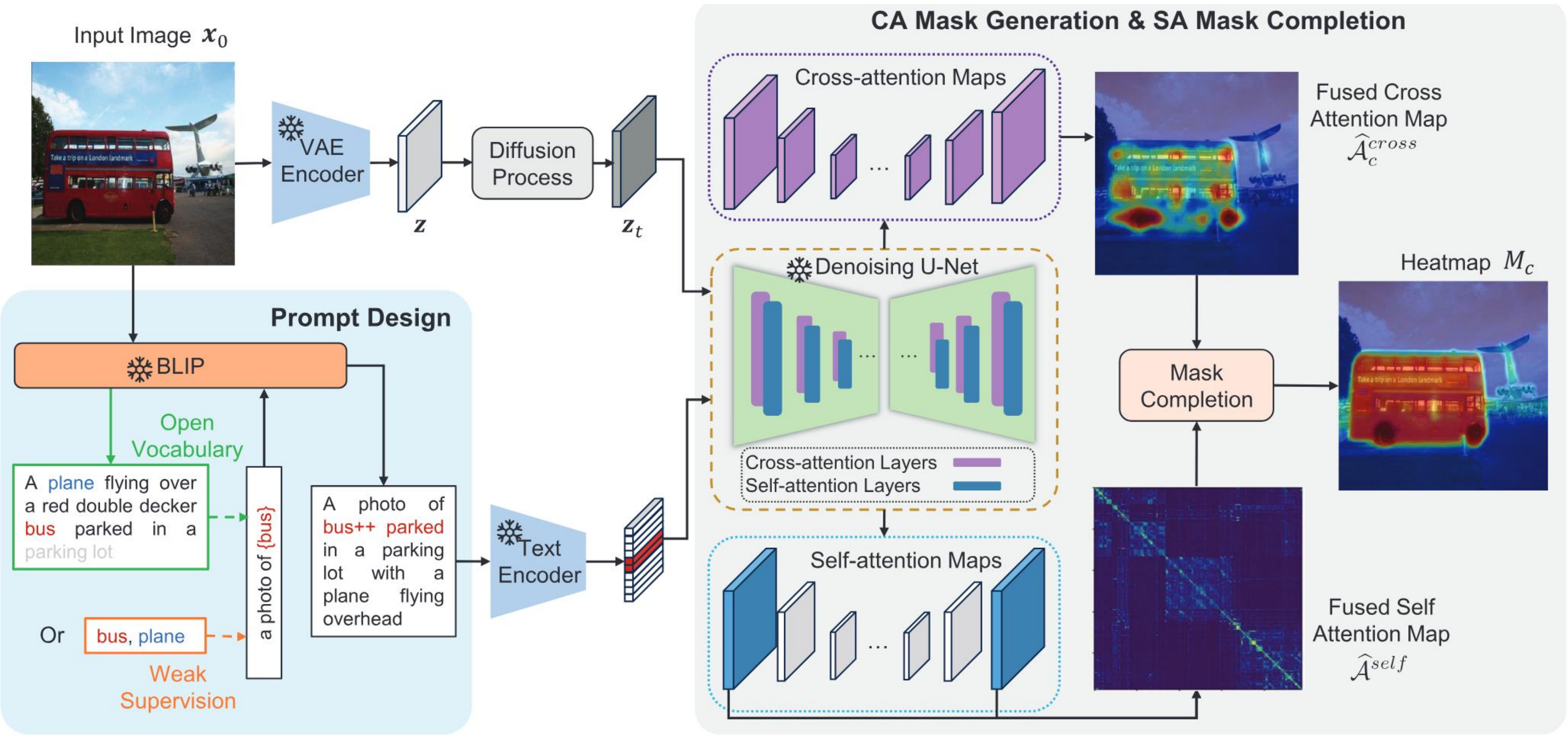
CLIP based: MaskCLIP/MaskCLIP+



Overview of MaskCLIP/MaskCLIP+. Compared to the conventional finetuning method, the key to the success of MaskCLIP is keeping the pre-trained weights frozen and making minimal adaptation to preserve the visual-language association. Besides, to compensate for the weakness of using the CLIP image encoder for segmentation, which is designed for classification, MaskCLIP+ uses the outputs of MaskCLIP as pseudo labels and trains a more advanced segmentation network such as DeepLabv2.

Zhou et al., Extract Free Dense Labels from CLIP, ECCV 2022

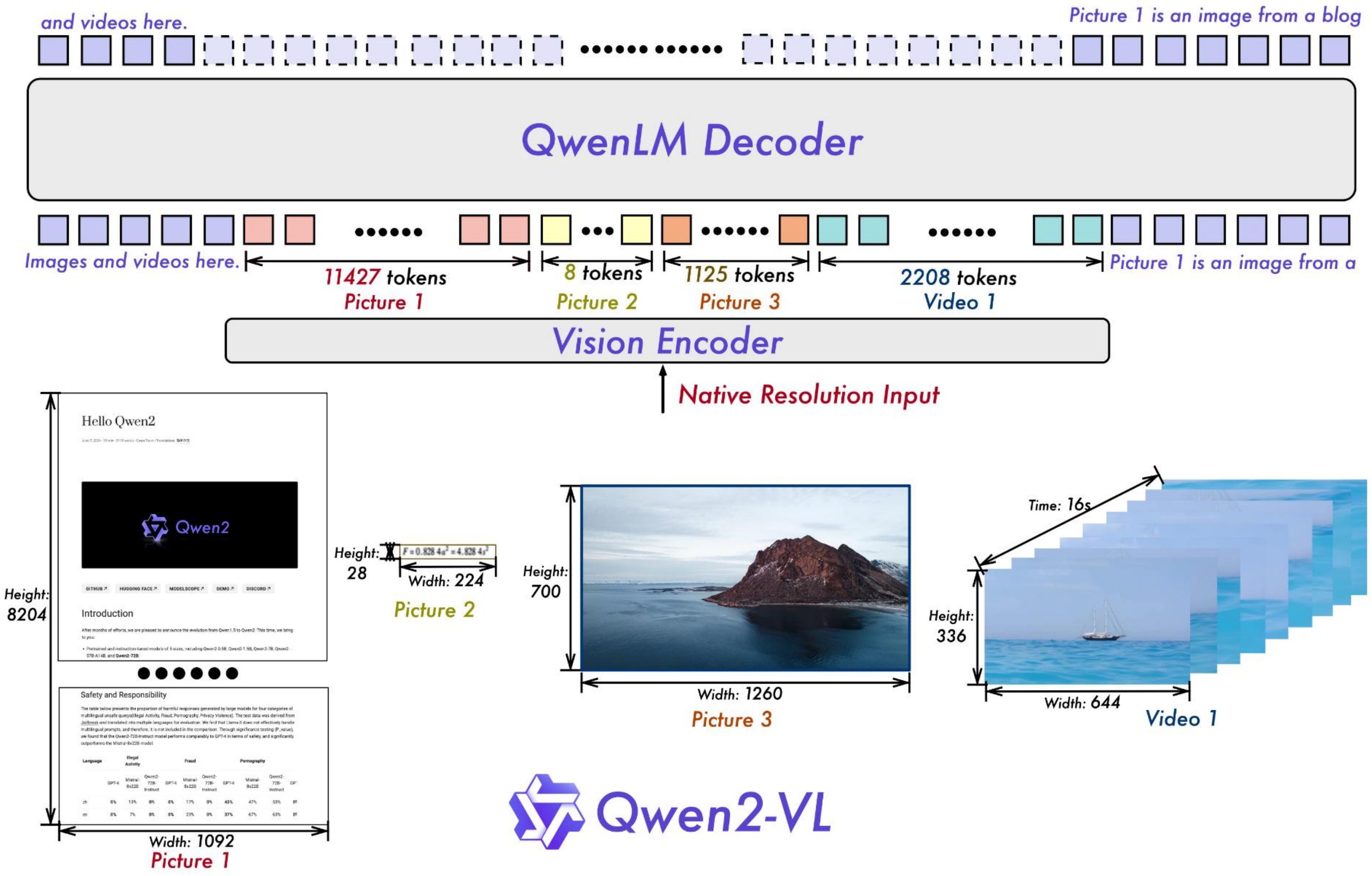
Diffusion based: DiffSegmenter

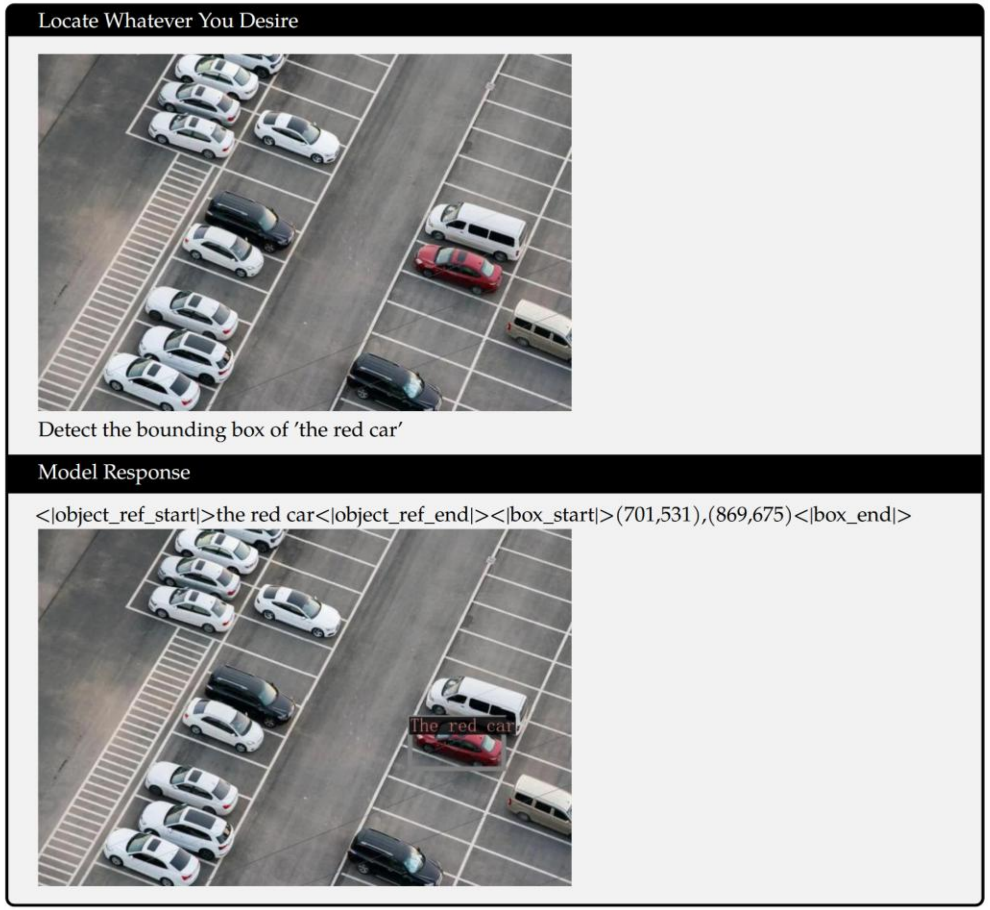


Overall framework of DiffSegmenter.

Wang et al., Diffusion Model is Secretly a Training-Free Open Vocabulary Semantic Segmenter, TIP 2025

Preliminary: MLLM

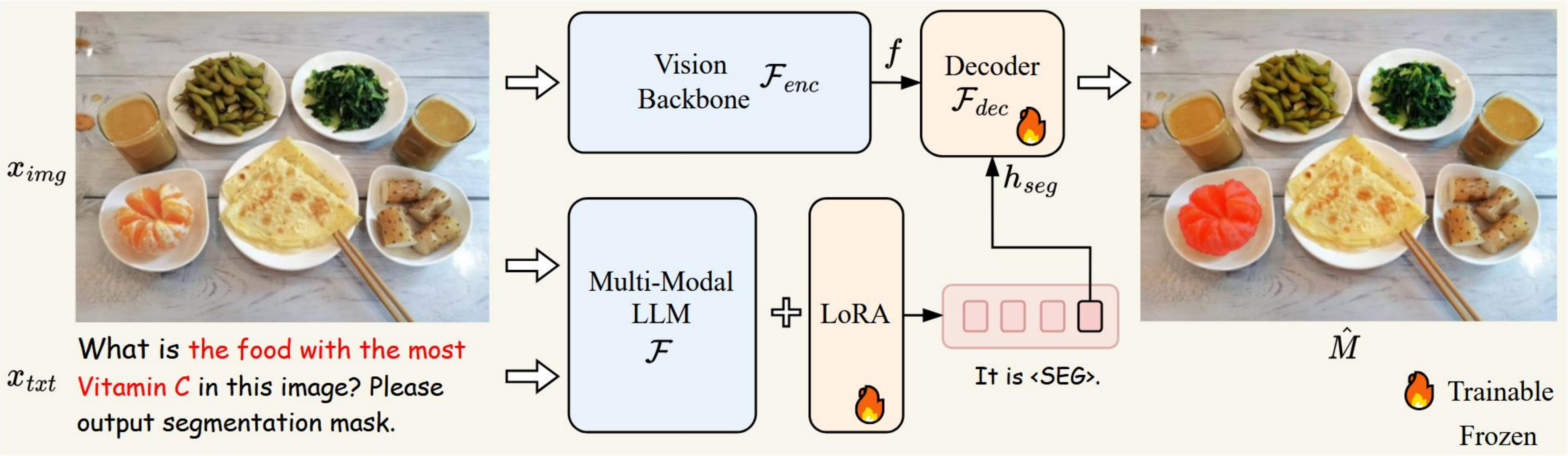




Left: Overview of Qwen2-VL. It is composed of a vision encoder for visual feature extraction, a LLM for language reasoning, and a connector bridging vision encoder and LLM. Right: Demo of visual grounding of Qwen2-VL in zero-shot manner.

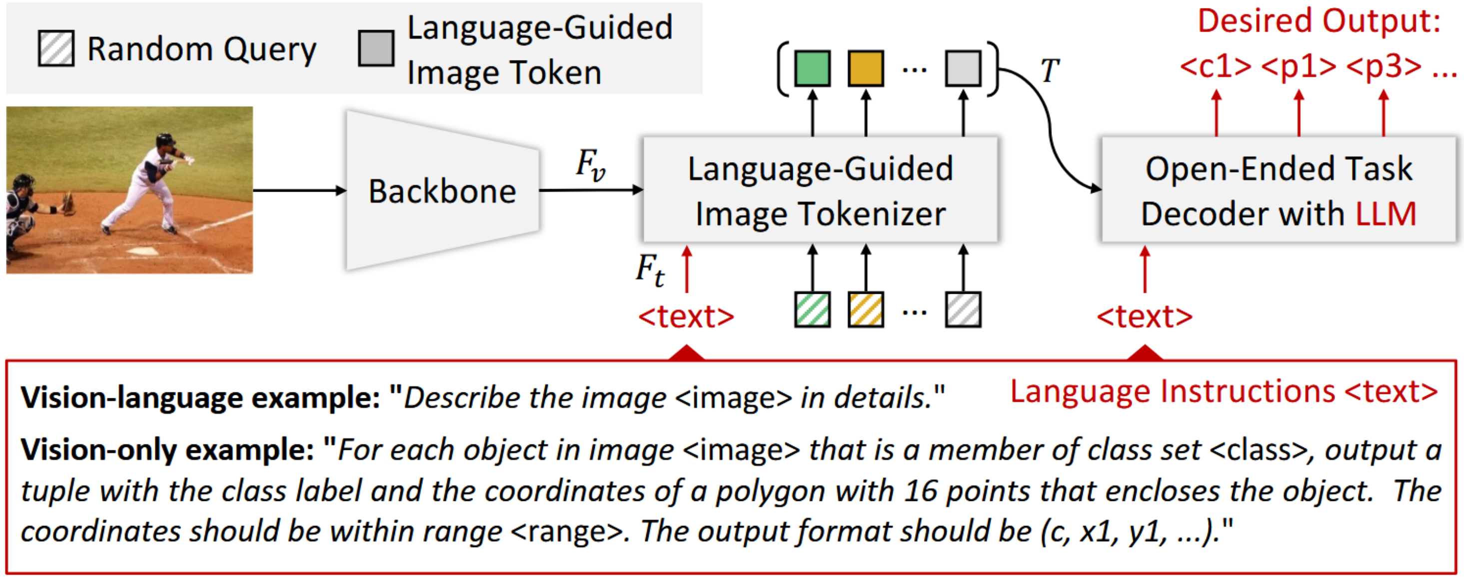
Wang et al., Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution, arXiv 2024

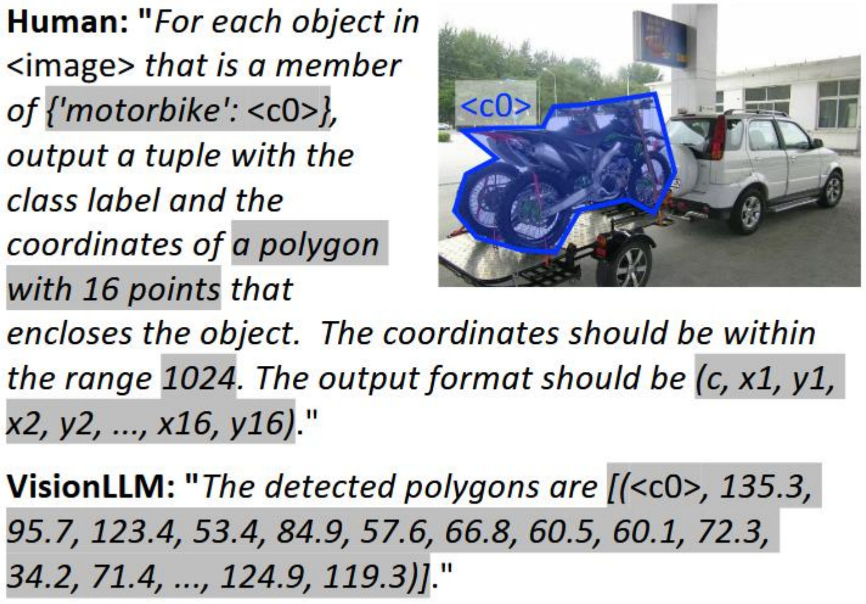
MLLM based: LISA



Overview of LISA. LISA adopt LLaVA-7B/13b as the base MLLM $\mathcal{F}$ along with ViT-H SAM backbone as the vision backbone $\mathcal{F}_{enc}$.

MLLM based: VisionLLM

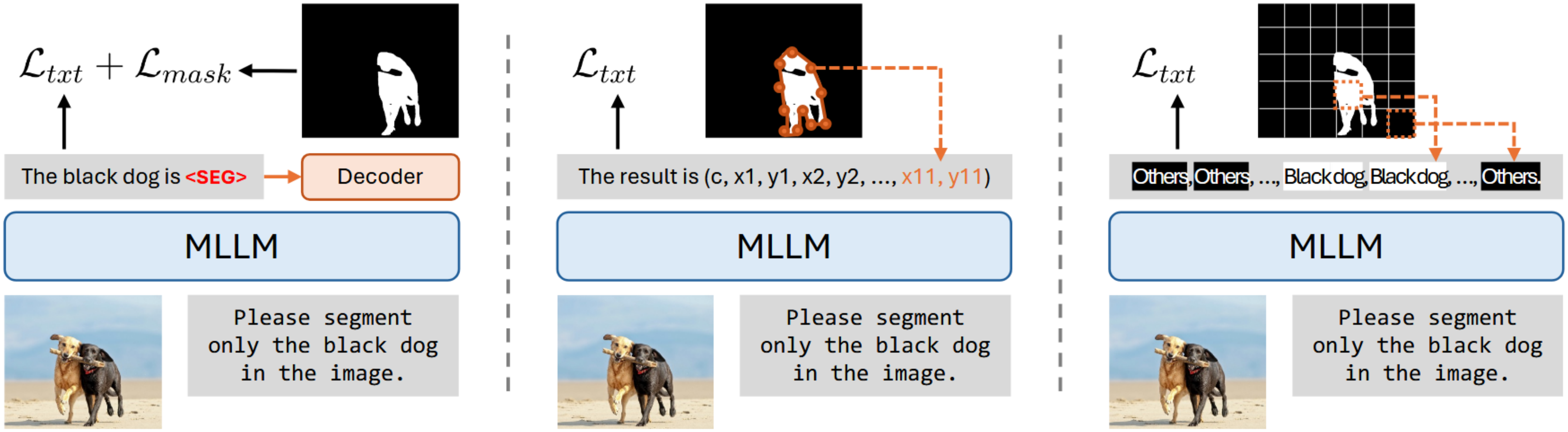




Left: Overview of VisionLLM. It consists of three parts: a unified language instruction designed to accommodate both vision and vision-language tasks, an image tokenizer that encodes visual information guided by language instructions, and an LLM-based open-ended task decoder that executes diverse tasks defined by language instructions. Right: Demo of referring segmentation.

Wang et al., VisionLLM: Large Language Model is also an Open-Ended Decoder for Vision-Centric Tasks, NeurIPS 2023

MLLM based: Text4Seg/Text4Seg++

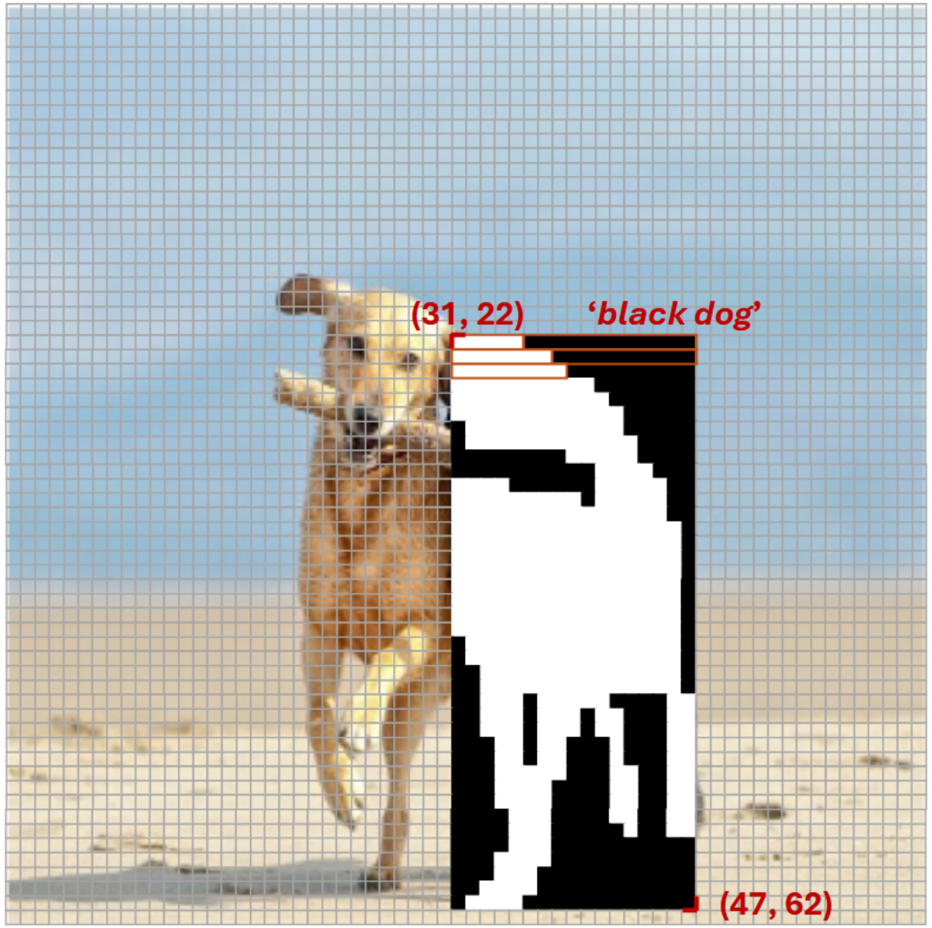


(a) *embeddings-as-mask* (b) *polygon coordinates* (c) *text-as-mask (ours)*

Different paradigms of MLLMs based image segmentation: (a) embeddings-as-mask paradigm that relies on additional segmentation decoder and loss (e.g., LISA); (b) polygon coordinates for image segmentation (e.g., VisionLLM); (c) Text4Seg++ text-as-mask paradigm that relies on semantically consistent text sequences.

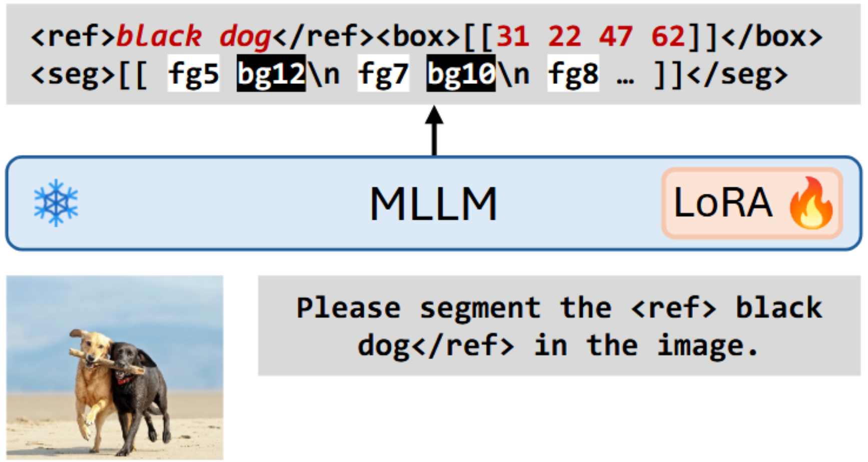
Lan et al., Text4Seg: Reimagining Image Segmentation as Text Generation, ICLR 2025
Lan et al., Text4Seg++: Advancing Image Segmentation via Generative Language Modeling, TPAMI 2026

MLLM based: Text4Seg/Text4Seg++



(a) Box-wise semantic descriptors

(b) Semantic bricks

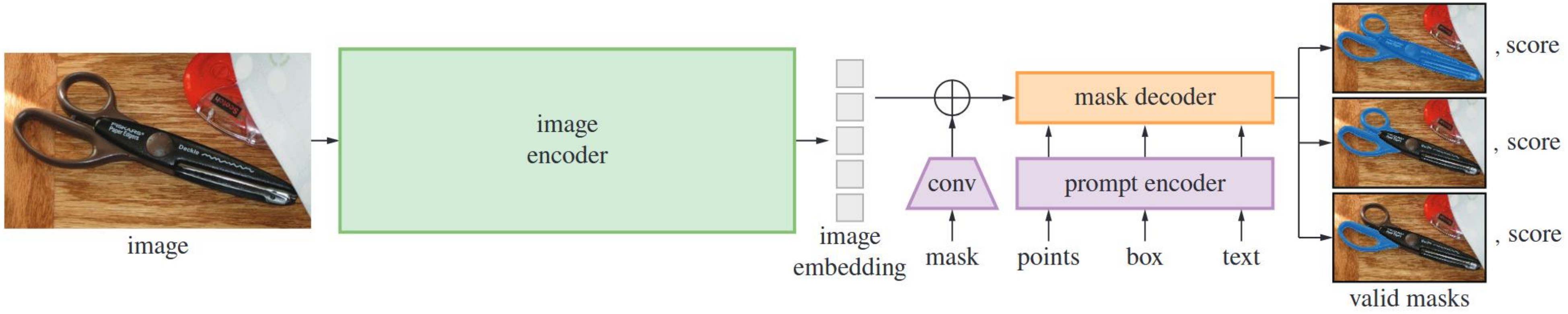


(c) Text4Seg++ architecture

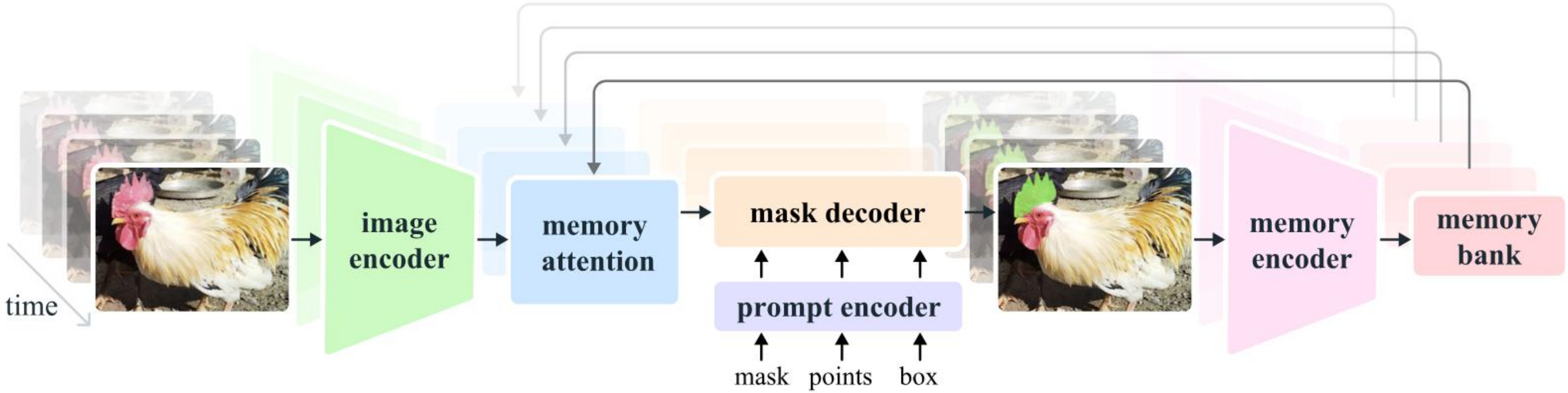
An illustration of (a) box-wise semantic descriptors for images, (b) semantic bricks and (c) Text4Seg++ framework.

Lan et al., Text4Seg: Reimagining Image Segmentation as Text Generation, ICLR 2025
Lan et al., Text4Seg++: Advancing Image Segmentation via Generative Language Modeling, TPAMI 2026

SAM Family



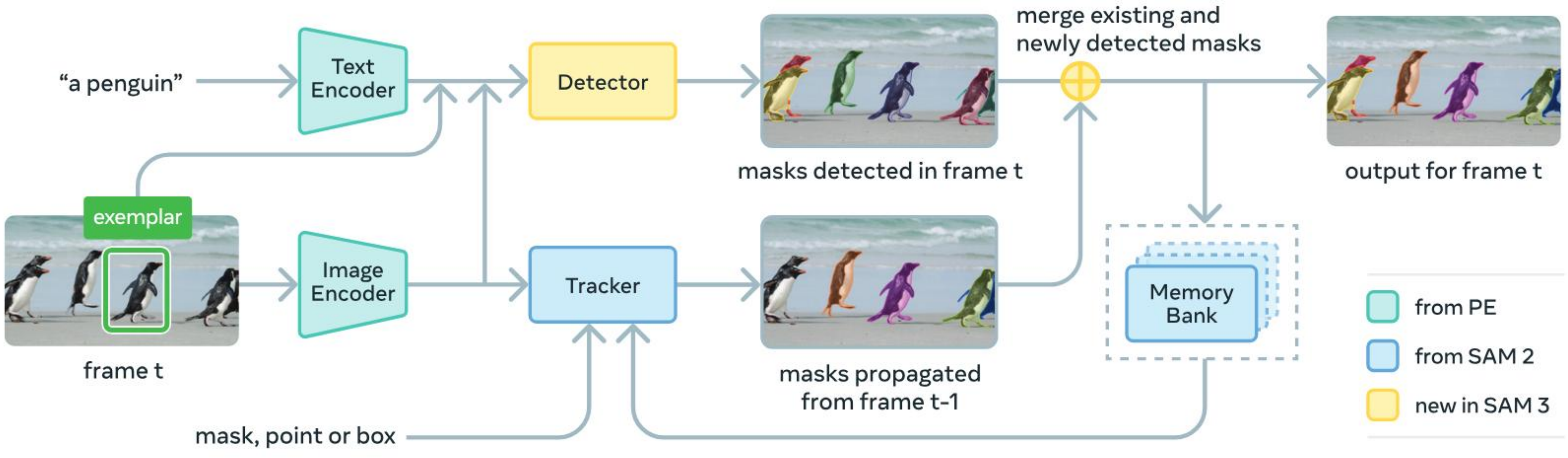
SAM architecture overview. First foundation model for segmentation specialized on interactive queries.



SAM 2 architecture overview. Extend SAM from image domain to video domain.

Kirillov et al., Segment Anything, ICCV 2023
Ravi et al., SAM 2: Segment Anything in Images and Videos, ICLR 2025
Carion et al., SAM 3: Segment Anything with Concepts, ICLR 2026

SAM Family

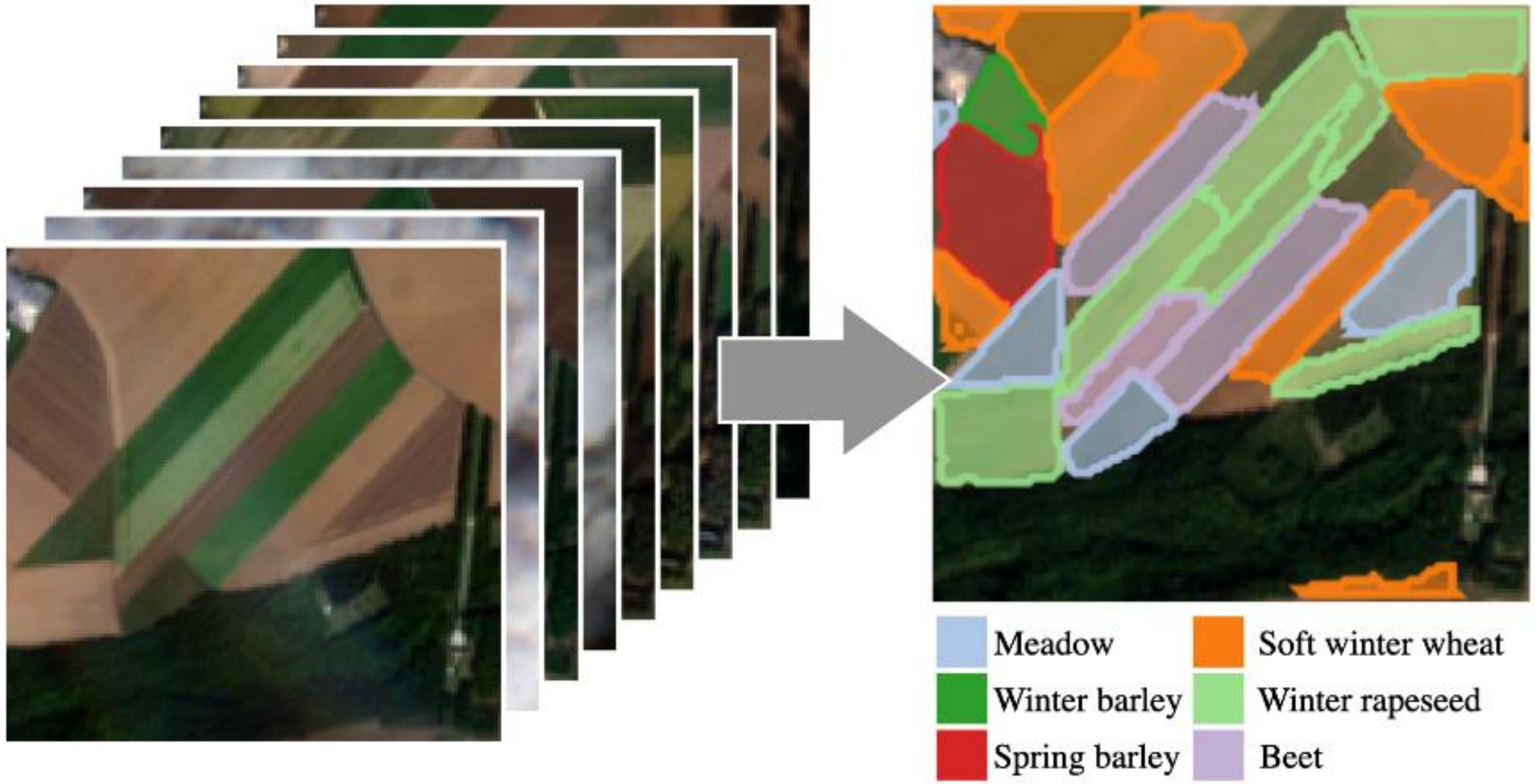


SAM 3 architecture overview. Extend SAM 2 to enable text prompt query for semantic segmentation.

Kirillov et al., Segment Anything, ICCV 2023
Ravi et al., SAM 2: Segment Anything in Images and Videos, ICLR 2025
Carion et al., SAM 3: Segment Anything with Concepts, ICLR 2026

Case Study in Cropland Recognition

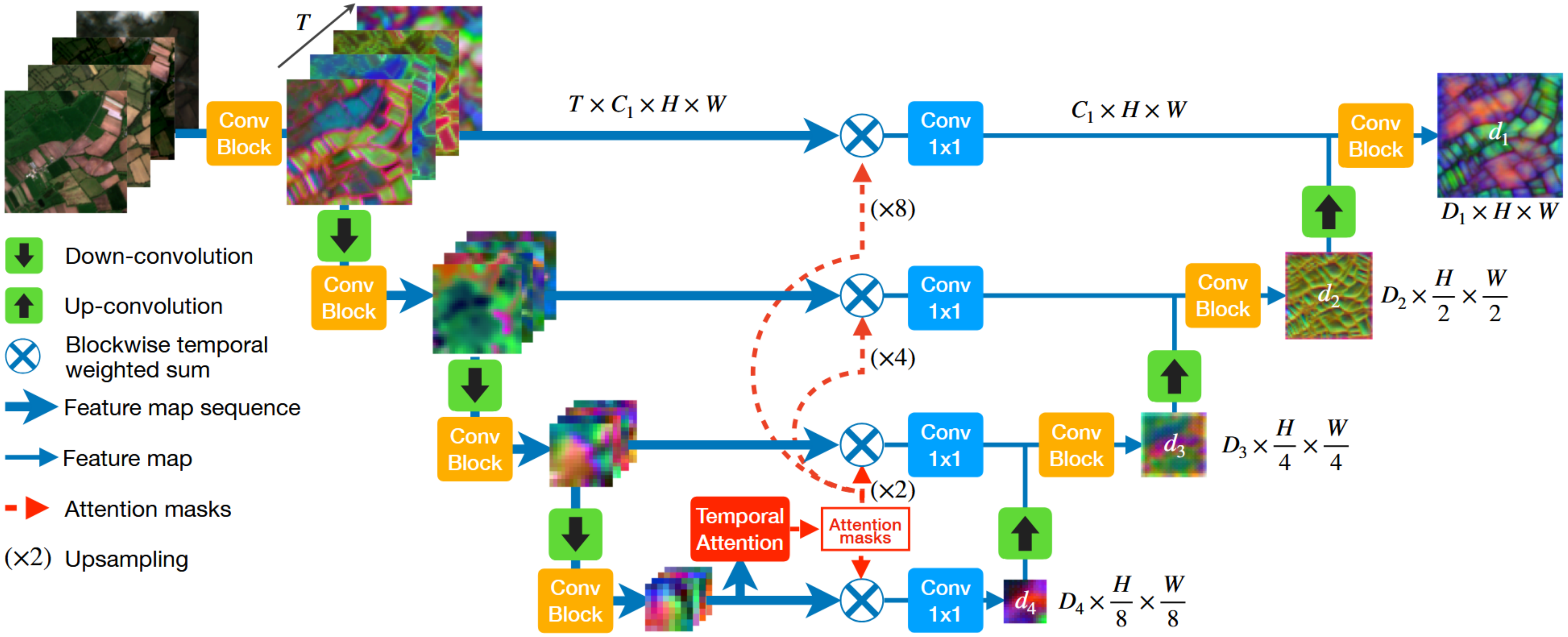
Data



PASTIS dataset. PASTIS is comprised of 2 433 sequences of multi-spectral images of shape 10 × 128 × 128. Each sequence contains between 38 and 61 observations taken between September 2018 and November 2019, for a total of over 2 billion pixels. The time between acquisitions is uneven with a median of 5 days.

Sainte Fare Garnot et al., Panoptic Segmentation of Satellite Image Time Series With Convolutional Temporal Attention Networks, ICCV 2021

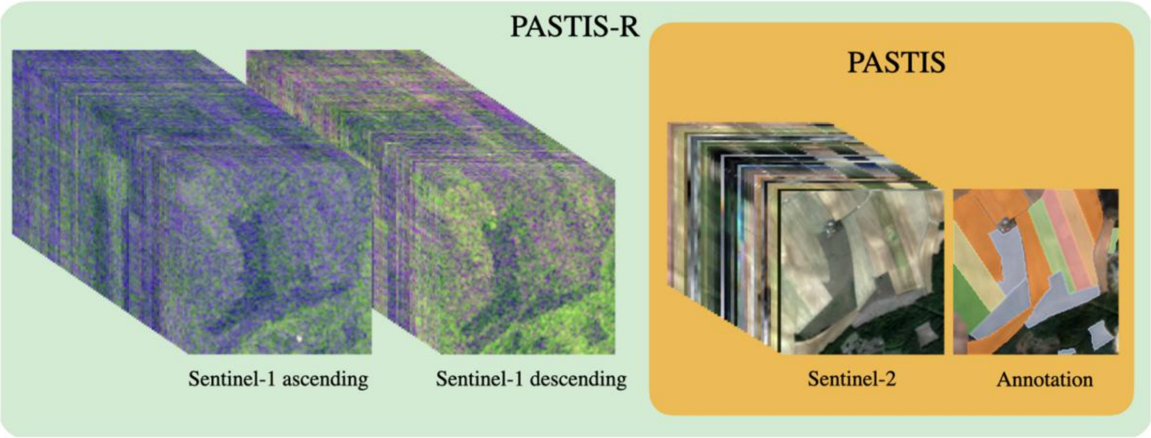
Framework



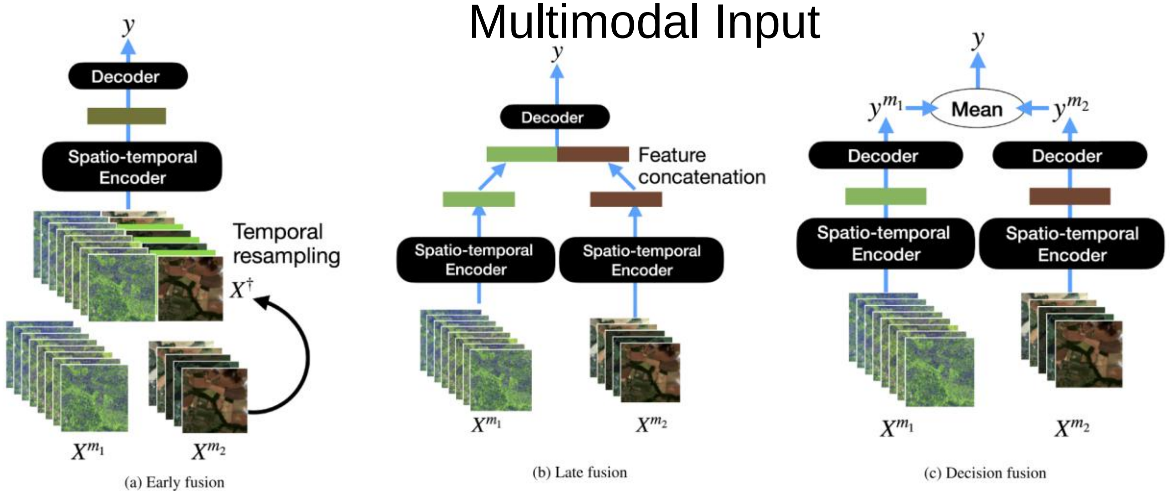
UNet with Temporal Attention Encoder (U-TAE). A sequence of images is processed in parallel by a shared convolutional encoder. At the lowest resolution, an attention-based temporal encoder produces a set of temporal attention masks for each pixel, which are then spatially interpolated at all resolutions. These masks are used to collapse the temporal dimension of the feature map sequences into a single map per resolution.

Sainte Fare Garnot et al., Panoptic Segmentation of Satellite Image Time Series With Convolutional Temporal Attention Networks, ICCV 2021

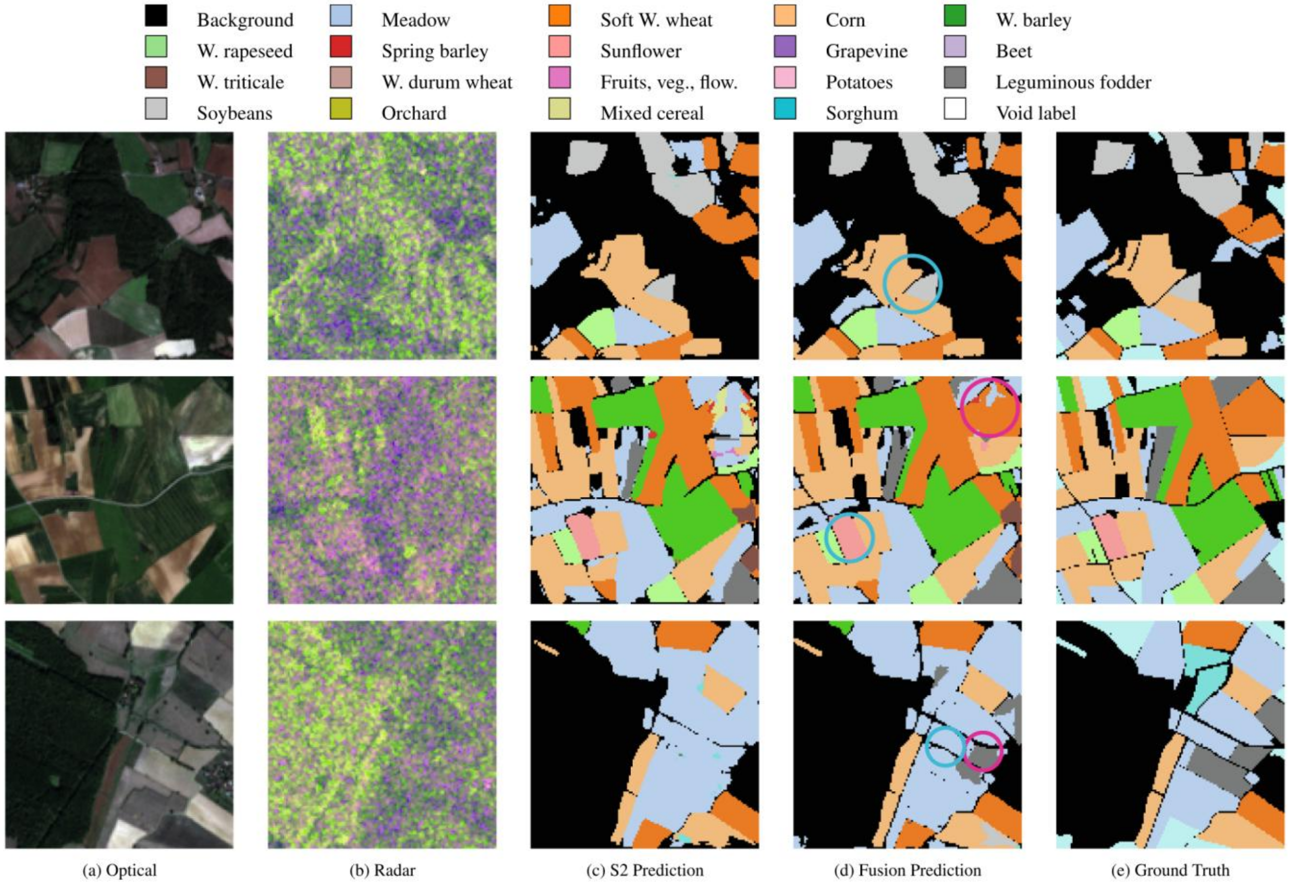
Multi-Modal Fusion



Multimodal Input



Fusion Schemas



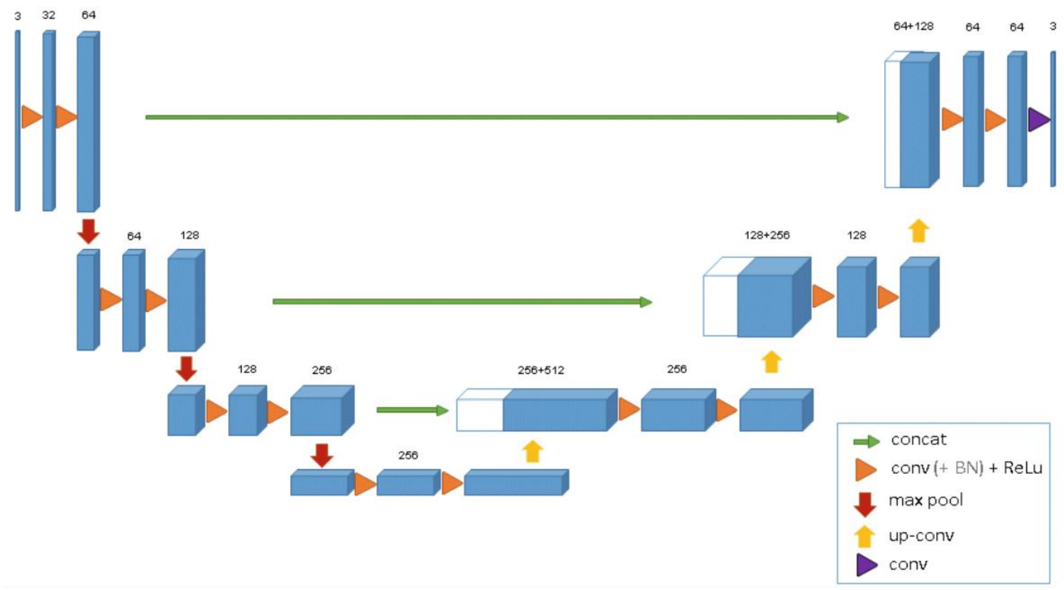
Qualitative Results for Semantic Segmentation.

Sainte Fare Garnot et al., Multi-modal temporal attention models for crop mapping from satellite time series, ISPRS 2022

Alternative Architecture Choice

Temporal Fusion!

2. SSL + Task Head: DINOv3 + DPT
3. Encoder + Decoder: Video Swin Transformer/ViViT + DPT
4. Encoder + Decoder: ResNet + UperNet



1. UNet 3D

Çiçek et al., 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation, MICCAI 2016

Bi-Temporal Semantic Change Detection

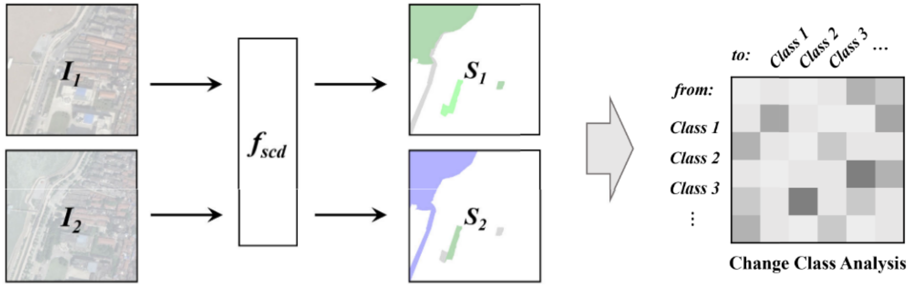
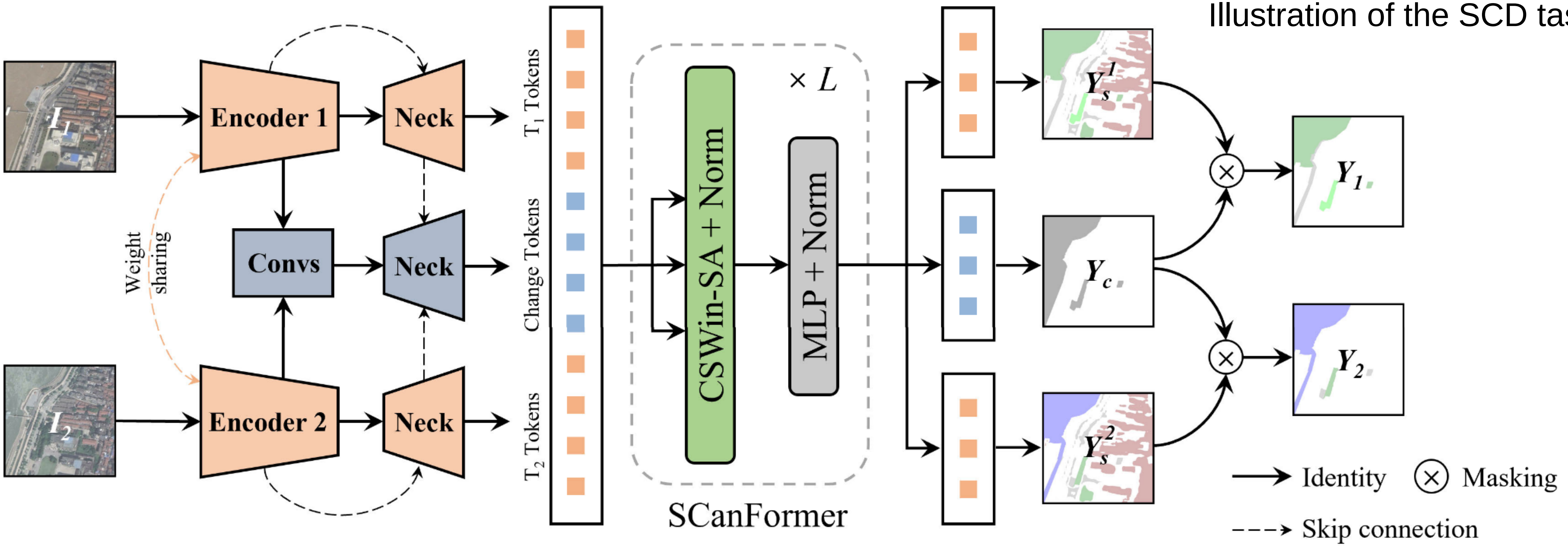


Illustration of the SCD task.



Architecture of the proposed ScanNet for semantic change detection (SCD).

Ding et al., Joint Spatio-Temporal Modeling for Semantic Change Detection in Remote Sensing Images, TGRS 2024